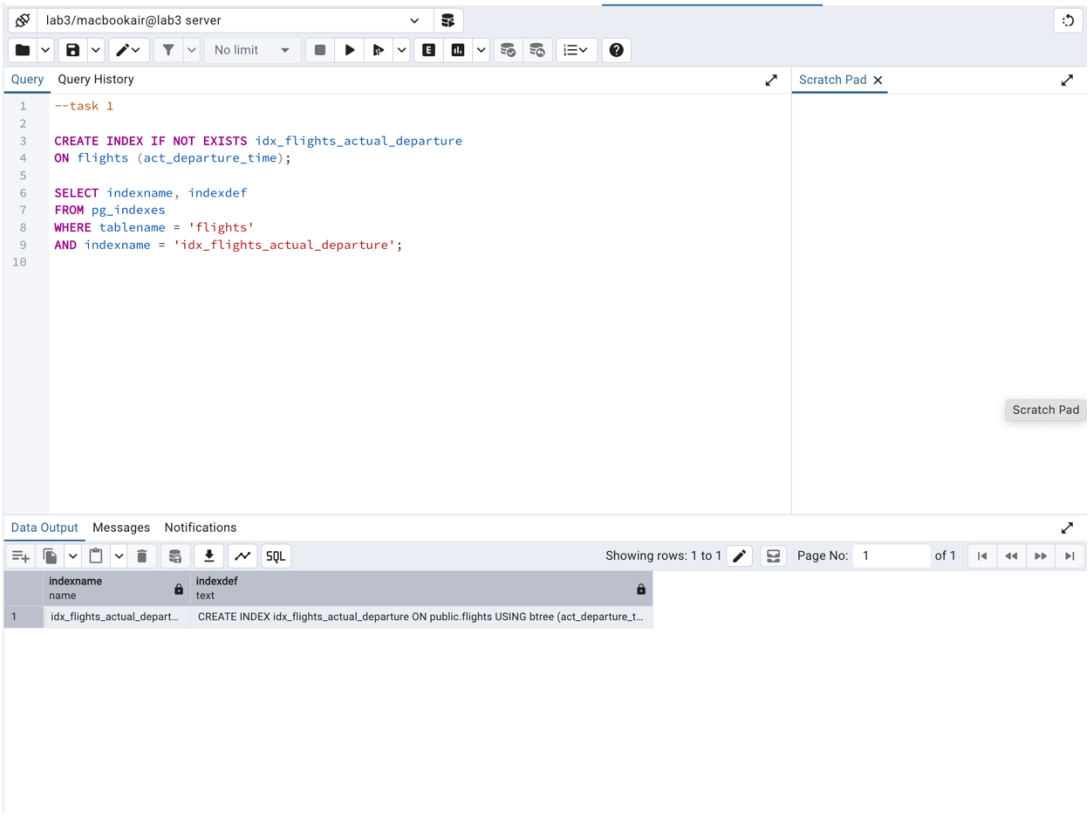
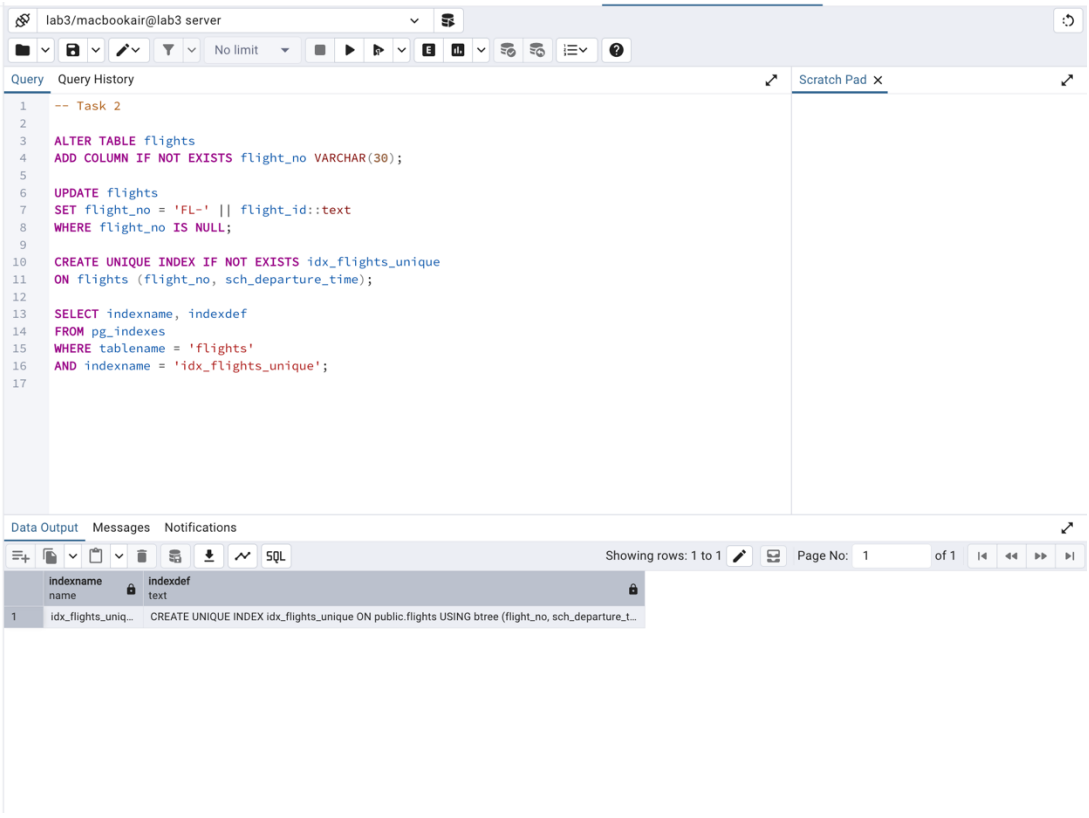
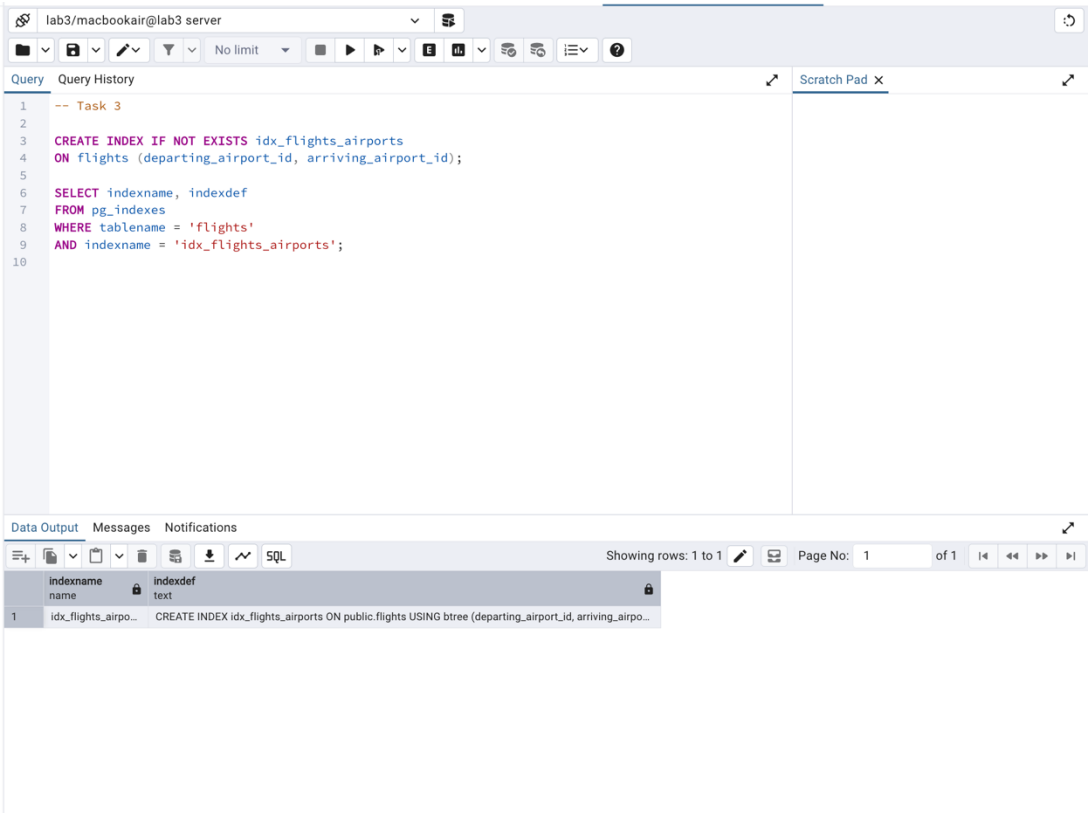
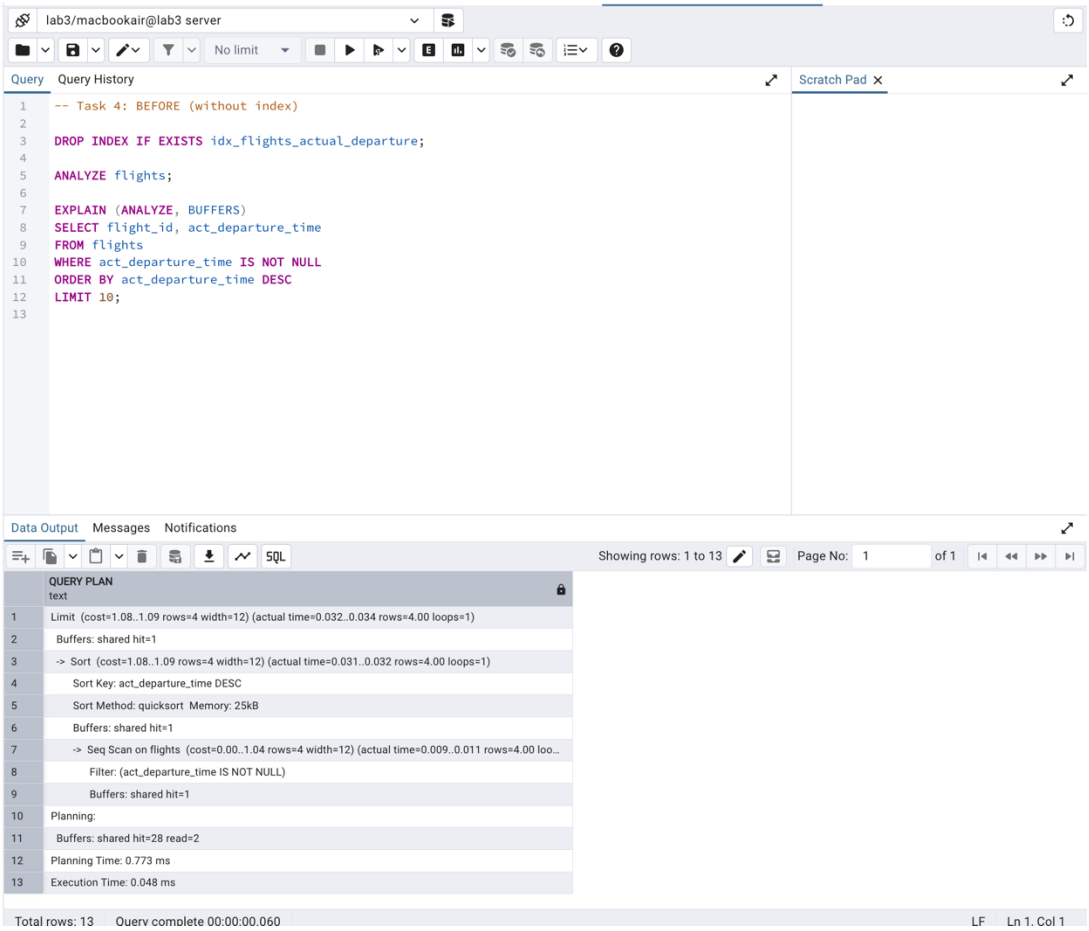
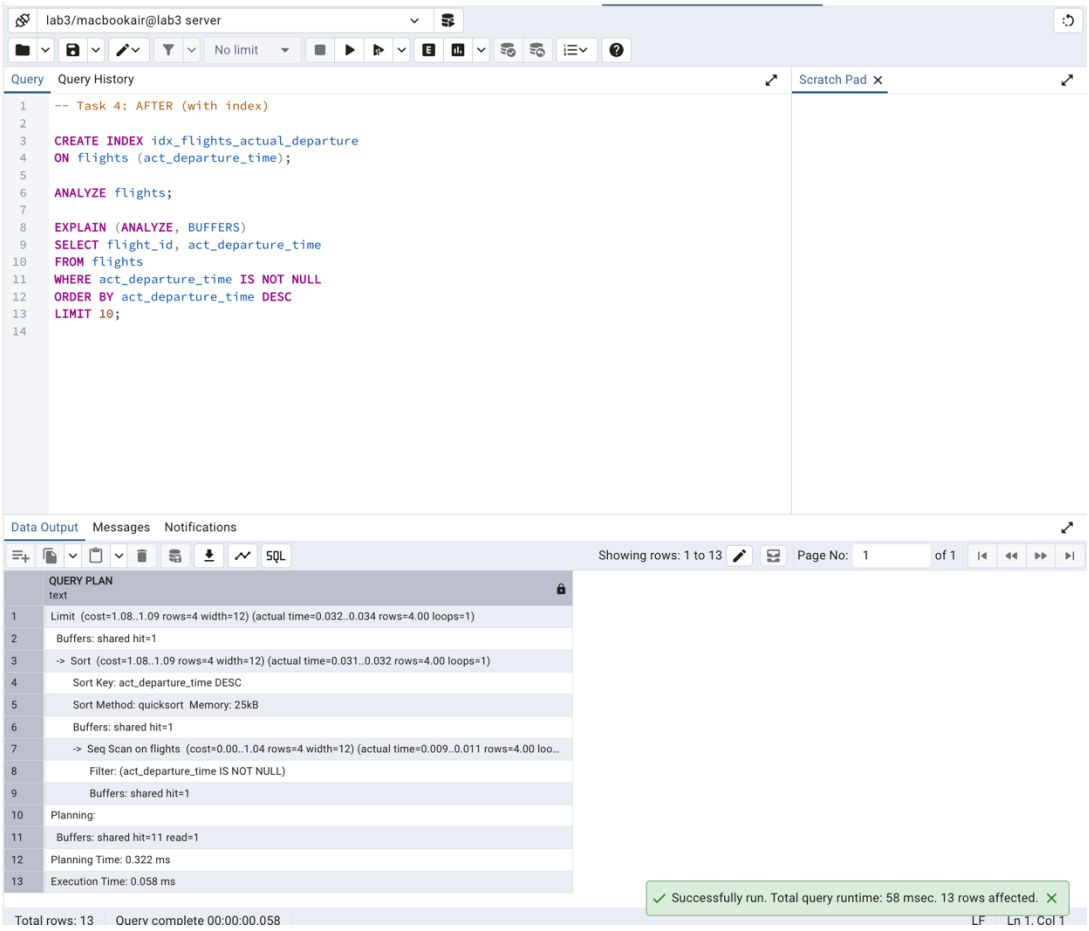
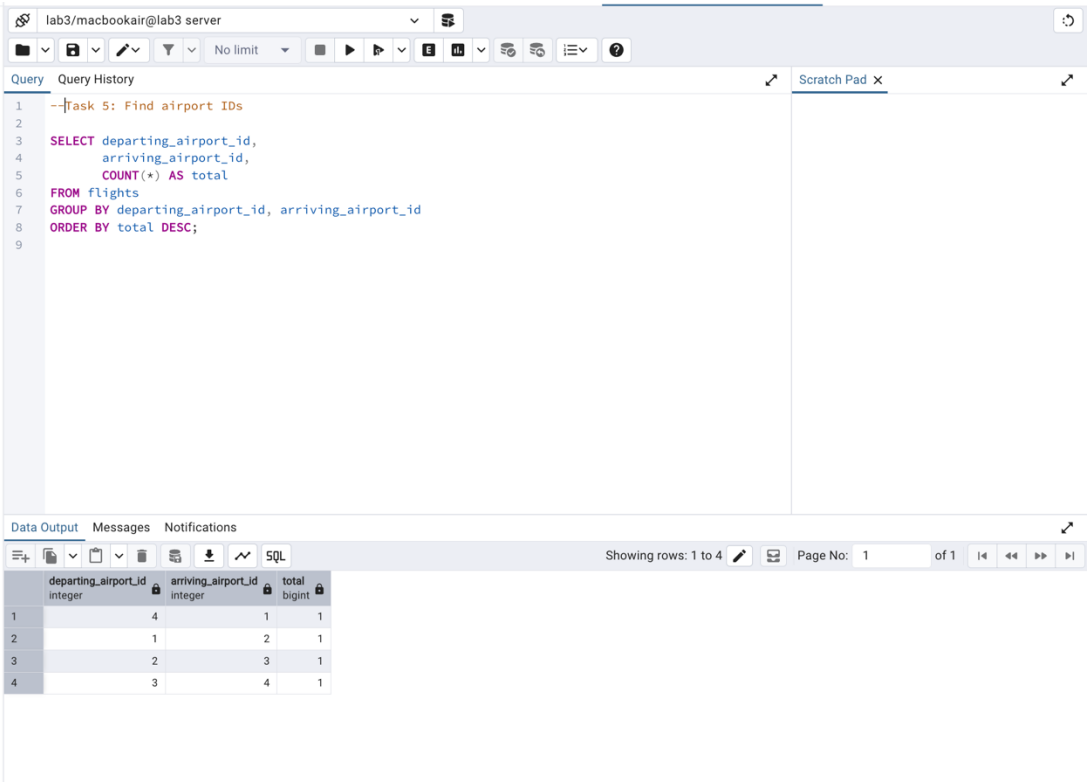

lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
1 --task 1
2
3 CREATE INDEX IF NOT EXISTS idx_flights_actual_departure
4 ON flights (act_departure_time);
5
6 SELECT indexname, indexdef
7 FROM pg_indexes
8 WHERE tablename = 'flights'
9 AND indexname = 'idx_flights_actual_departure';
10
Scratch Pad
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 1
Page No: 1
of 1
indexname
name
indexdef
text
1
idx_flights_actual_depart...
CREATE INDEX idx_flights_actual_departure ON public.flights USING btree (act_departure_t...

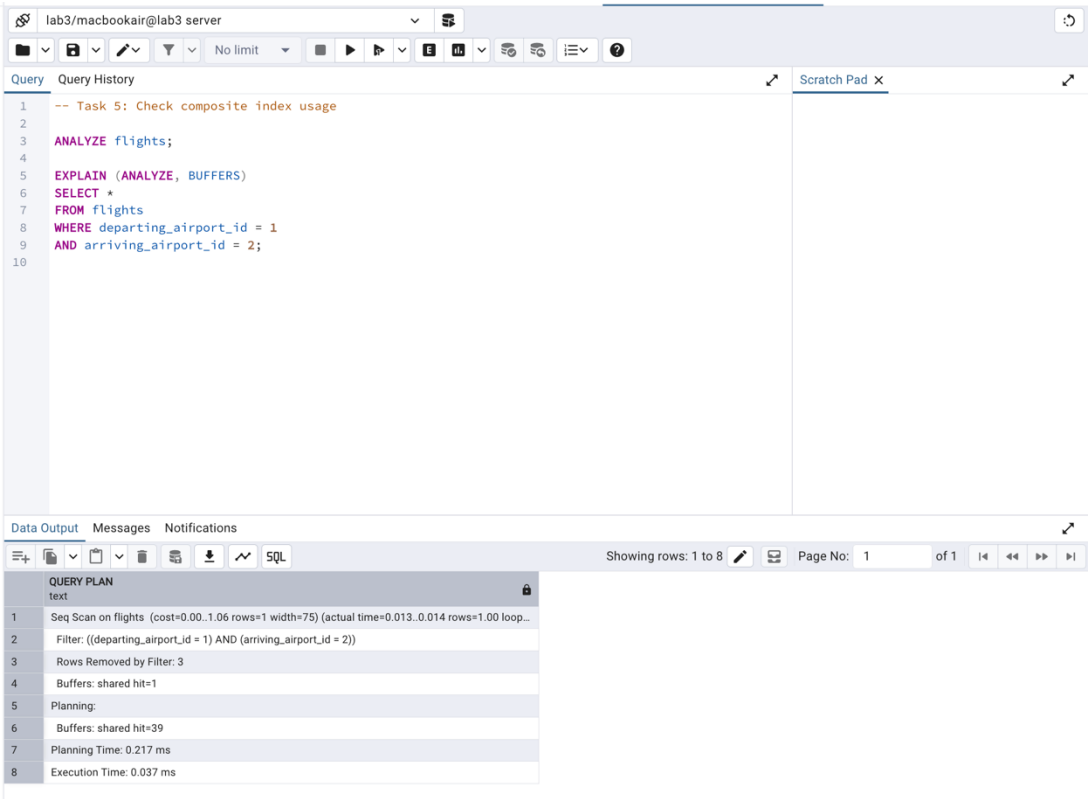

lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
1 -- Task 2
2
3 ALTER TABLE flights
4 ADD COLUMN IF NOT EXISTS flight_no VARCHAR(30);
5
6 UPDATE flights
7 SET flight_no = 'FL-' || flight_id::text
8 WHERE flight_no IS NULL;
9
10 CREATE UNIQUE INDEX IF NOT EXISTS idx_flights_unique
11 ON flights (flight_no, sch_departure_time);
12
13 SELECT indexname, indexdef
14 FROM pg_indexes
15 WHERE tablename = 'flights'
16 AND indexname = 'idx_flights_unique';
17
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 1
Page No: 1
of 1
indexname
name
indexdef
text
1
idx_flights_uniq...
CREATE UNIQUE INDEX idx_flights_unique ON public.flights USING btree (flight_no, sch_departure_t...

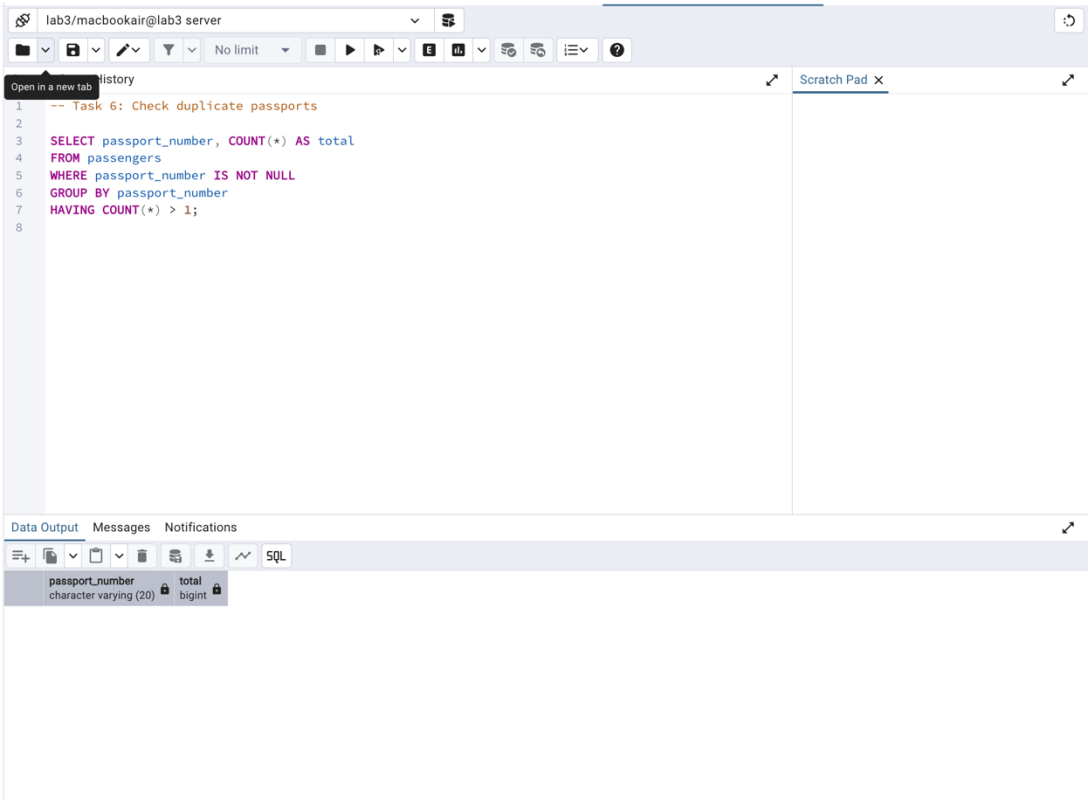

lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 3
CREATE INDEX IF NOT EXISTS idx_flights_airports
ON flights (departing_airport_id, arriving_airport_id);
SELECT indexname, indexdef
FROM pg_indexes
WHERE tablename = 'flights'
AND indexname = 'idx_flights_airports';
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 1
Page No: 1
of 1
indexname
name
indexdef
text
1
idx_flights_airpo...
CREATE INDEX idx_flights_airports ON public.flights USING btree (departing_airport_id, arriving_airpo...

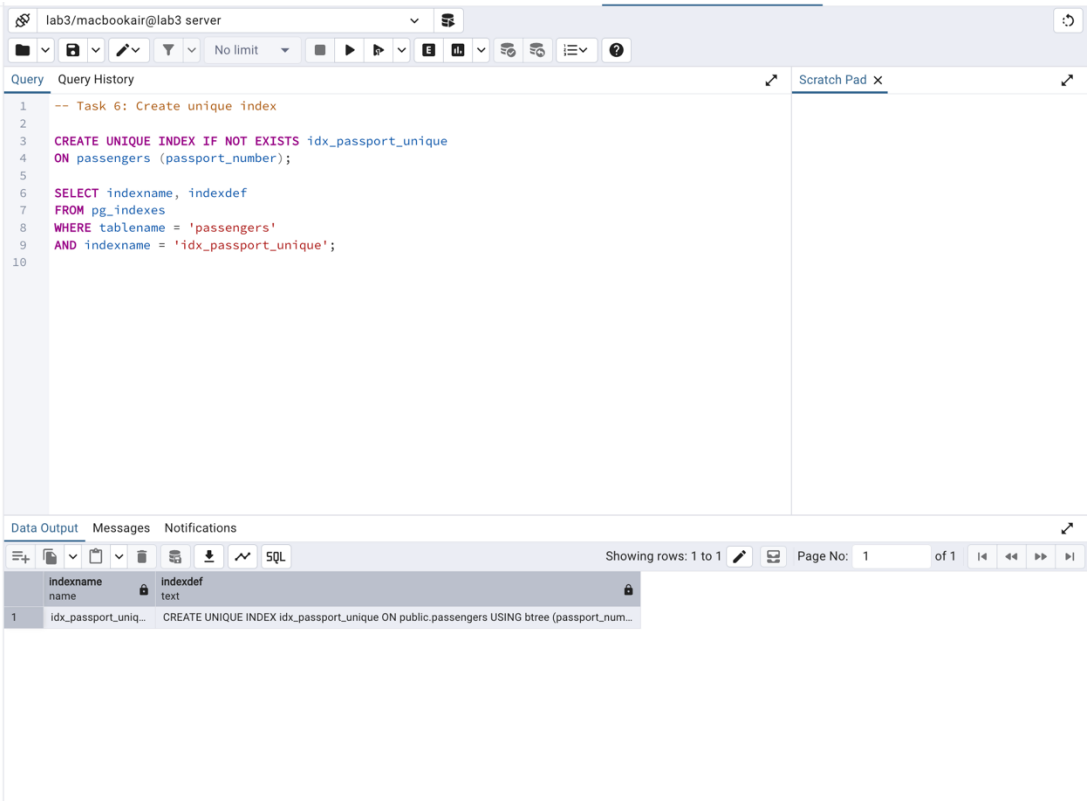

lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 4: BEFORE (without index)
DROP INDEX IF EXISTS idx_flights_actual_departure;
ANALYZE flights;
EXPLAIN (ANALYZE, BUFFERS)
SELECT flight_id, act_departure_time
FROM flights
WHERE act_departure_time IS NOT NULL
ORDER BY act_departure_time DESC
LIMIT 10;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 13
Page No: 1
of 1
QUERY PLAN
text
1 Limit (cost=1.08..1.09 rows=4 width=12) (actual time=0.032..0.034 rows=4.00 loops=1)
2 Buffers: shared hit=1
3 -> Sort (cost=1.08..1.09 rows=4 width=12) (actual time=0.031..0.032 rows=4.00 loops=1)
4 Sort Key: act_departure_time DESC
5 Sort Method: quicksort Memory: 25kB
6 Buffers: shared hit=1
7 -> Seq Scan on flights (cost=0.00..1.04 rows=4 width=12) (actual time=0.009..0.011 rows=4.00 loo...
8 Filter: (act_departure_time IS NOT NULL)
9 Buffers: shared hit=1
10 Planning:
11 Buffers: shared hit=28 read=2
12 Planning Time: 0.773 ms
13 Execution Time: 0.048 ms
Total rows: 13
Query complete 00:00:00.060
LF
Ln 1, Col 1

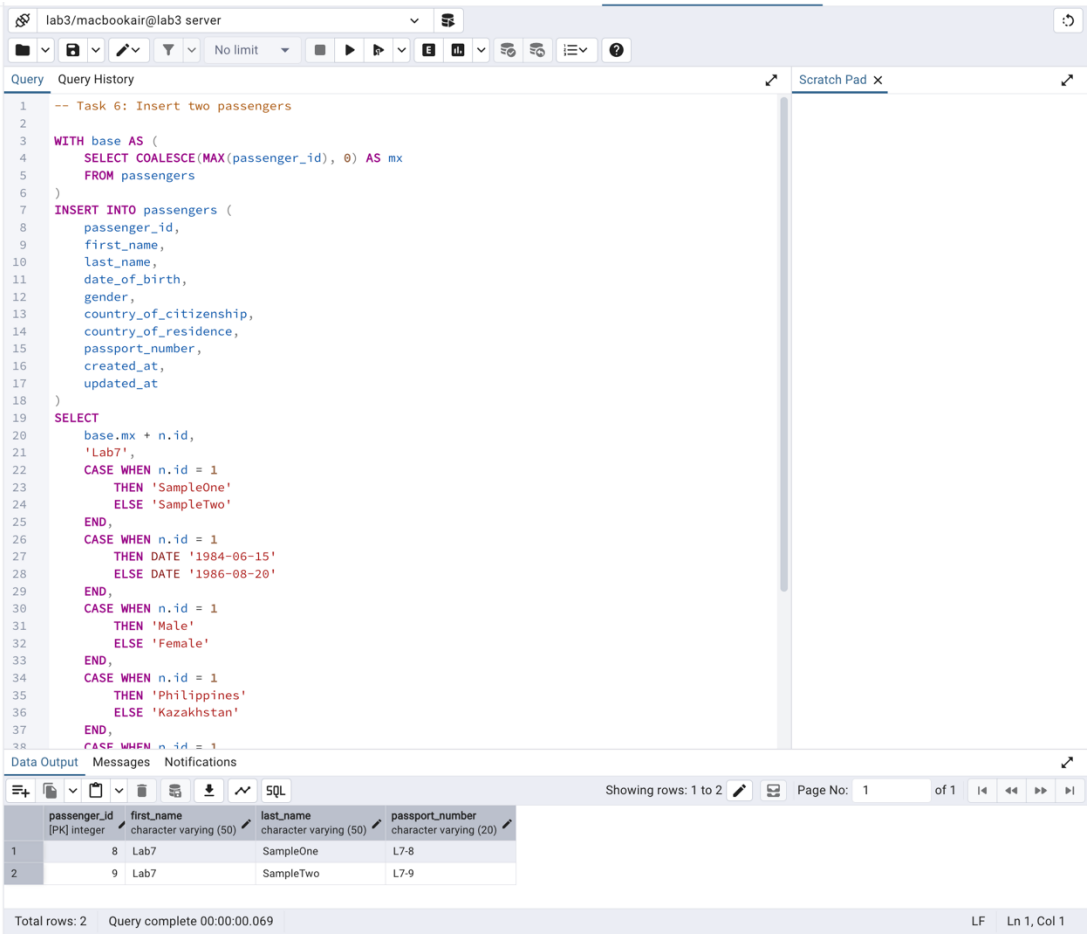

lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 4: AFTER (with index)
CREATE INDEX idx_flights_actual_departure
ON flights (act_departure_time);
ANALYZE flights;
EXPLAIN (ANALYZE, BUFFERS)
SELECT flight_id, act_departure_time
FROM flights
WHERE act_departure_time IS NOT NULL
ORDER BY act_departure_time DESC
LIMIT 10;
Data Output
Messages
Notifications
Showing rows: 1 to 13
Page No: 1
of 1
QUERY PLAN
text
1 Limit (cost=1.08..1.09 rows=4 width=12) (actual time=0.032..0.034 rows=4.00 loops=1)
2 Buffers: shared hit=1
3 -> Sort (cost=1.08..1.09 rows=4 width=12) (actual time=0.031..0.032 rows=4.00 loops=1)
4 Sort Key: act_departure_time DESC
5 Sort Method: quicksort Memory: 25kB
6 Buffers: shared hit=1
7 -> Seq Scan on flights (cost=0.00..1.04 rows=4 width=12) (actual time=0.009..0.011 rows=4.00 loo...
8 Filter: (act_departure_time IS NOT NULL)
9 Buffers: shared hit=1
10 Planning:
11 Buffers: shared hit=11 read=1
12 Planning Time: 0.322 ms
13 Execution Time: 0.058 ms
Successfully run. Total query runtime: 58 msec. 13 rows affected.
Total rows: 13
Query complete 00:00:00.058
LF
Ln 1, Col 1

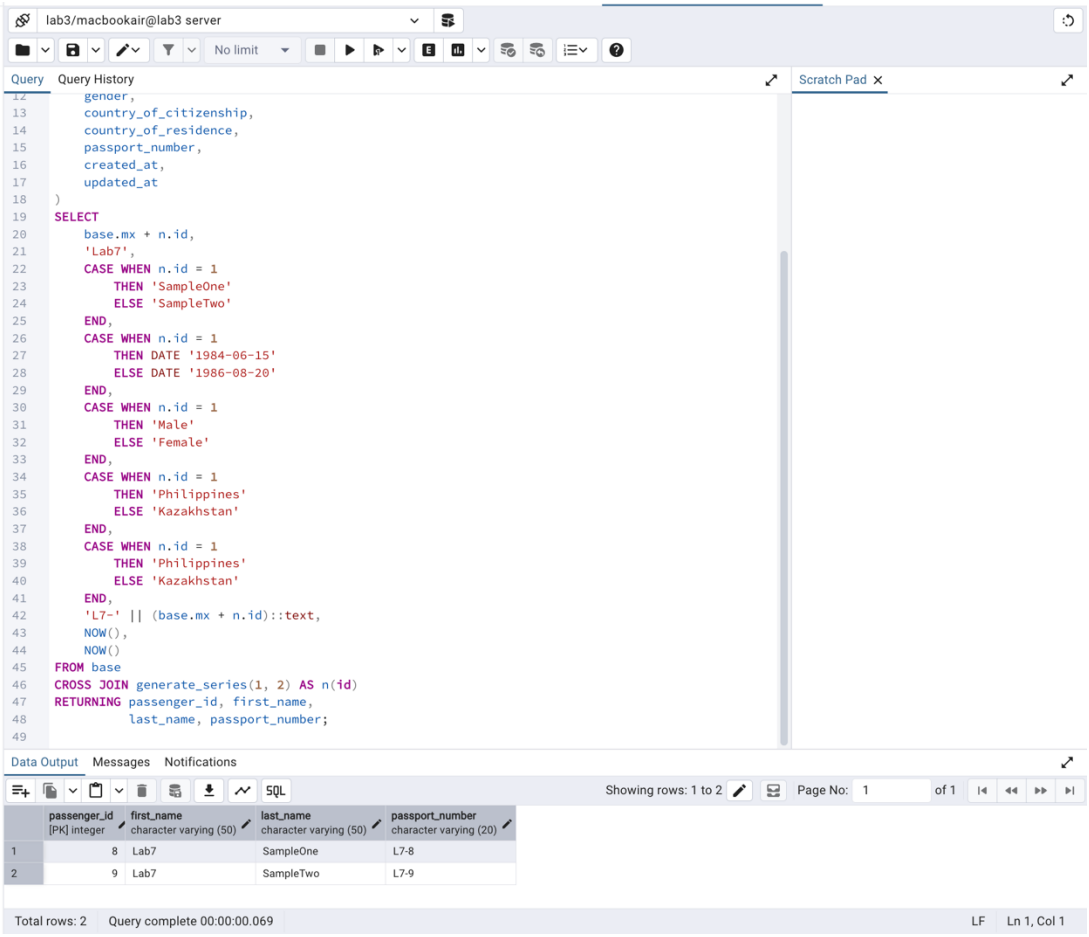

lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 5: Find airport IDs
SELECT departing_airport_id,
arriving_airport_id,
COUNT(*) AS total
FROM flights
GROUP BY departing_airport_id, arriving_airport_id
ORDER BY total DESC;
Data Output
Messages
Notifications
Showing rows: 1 to 4
Page No: 1
of 1
departing_airport_id integer
arriving_airport_id integer
total bigint
1 4 1 1
2 1 2 1
3 2 3 1
4 3 4 1

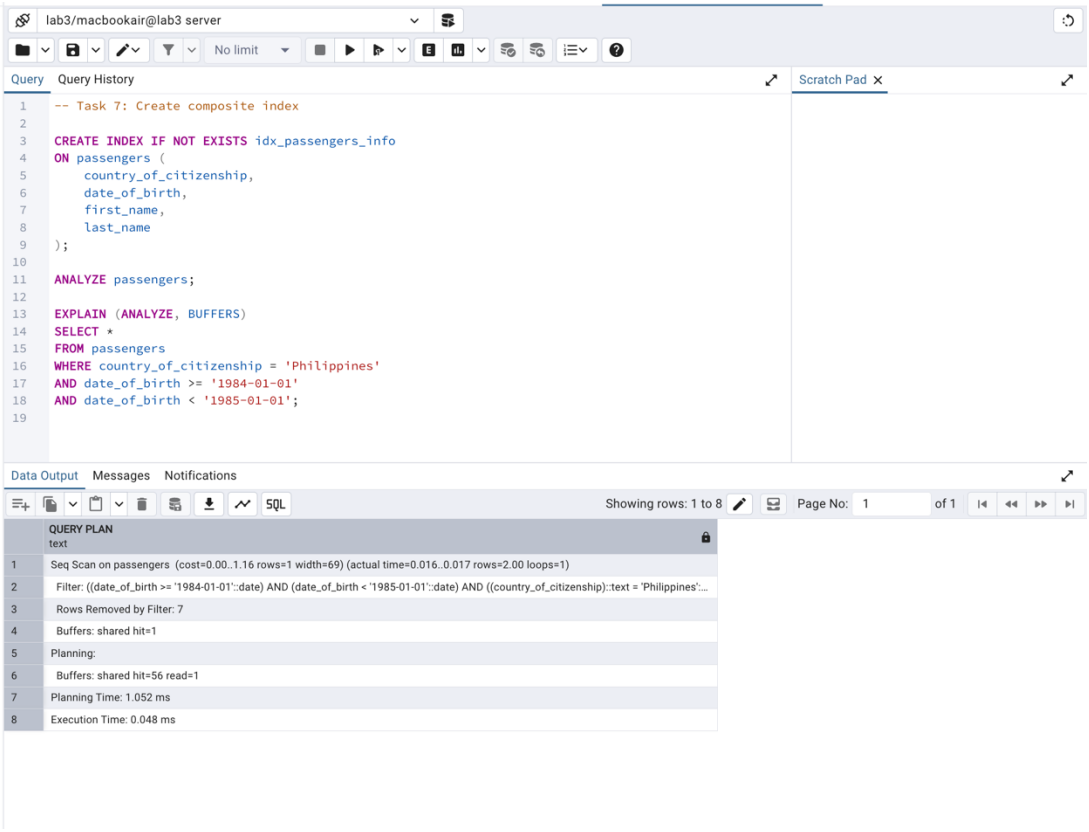

lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 5: Check composite index usage
ANALYZE flights;
EXPLAIN (ANALYZE, BUFFERS)
SELECT *
FROM flights
WHERE departing_airport_id = 1
AND arriving_airport_id = 2;
Data Output
Messages
Notifications
Showing rows: 1 to 8
Page No: 1
of 1
QUERY PLAN text
1 Seq Scan on flights (cost=0.00..1.06 rows=1 width=75) (actual time=0.013..0.014 rows=1 loop...
2 Filter: ((departing_airport_id = 1) AND (arriving_airport_id = 2))
3 Rows Removed by Filter: 3
4 Buffers: shared hit=1
5 Planning:
6 Buffers: shared hit=39
7 Planning Time: 0.217 ms
8 Execution Time: 0.037 ms

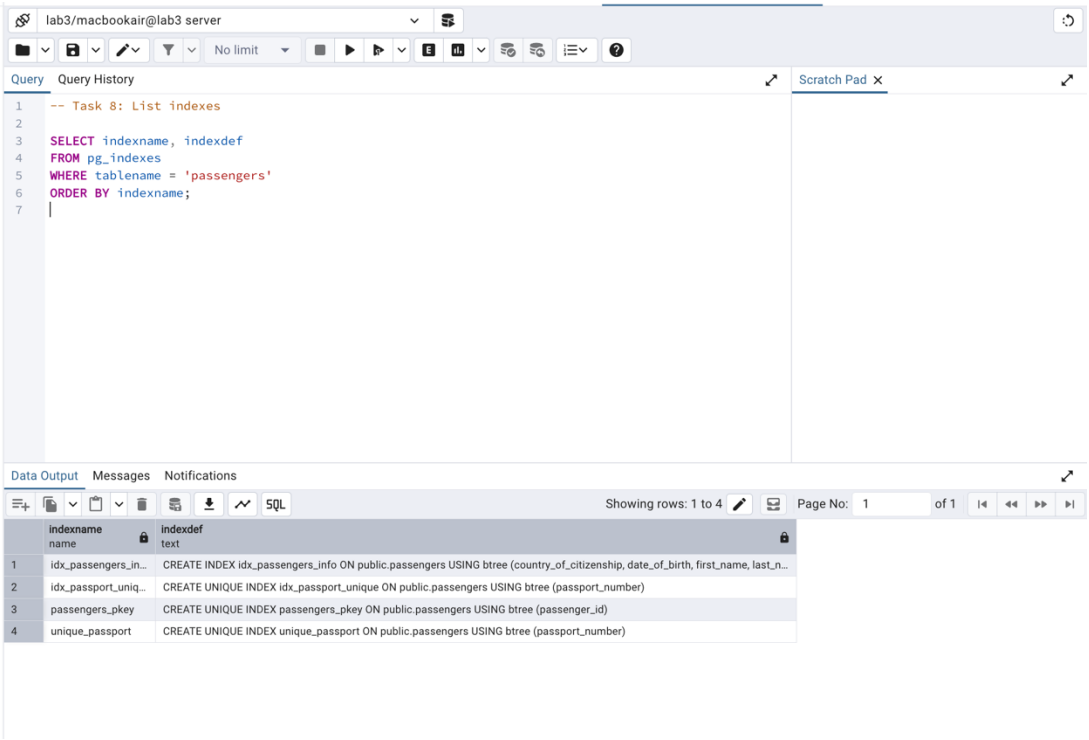

lab3/macbookair@lab3 server
No limit
Open in a new tab
Scratch Pad
-- Task 6: Check duplicate passports
SELECT passport_number, COUNT(*) AS total
FROM passengers
WHERE passport_number IS NOT NULL
GROUP BY passport_number
HAVING COUNT(*) > 1;
Data Output
Messages
Notifications
SQL
passport_number
character varying (20)
total
bigint

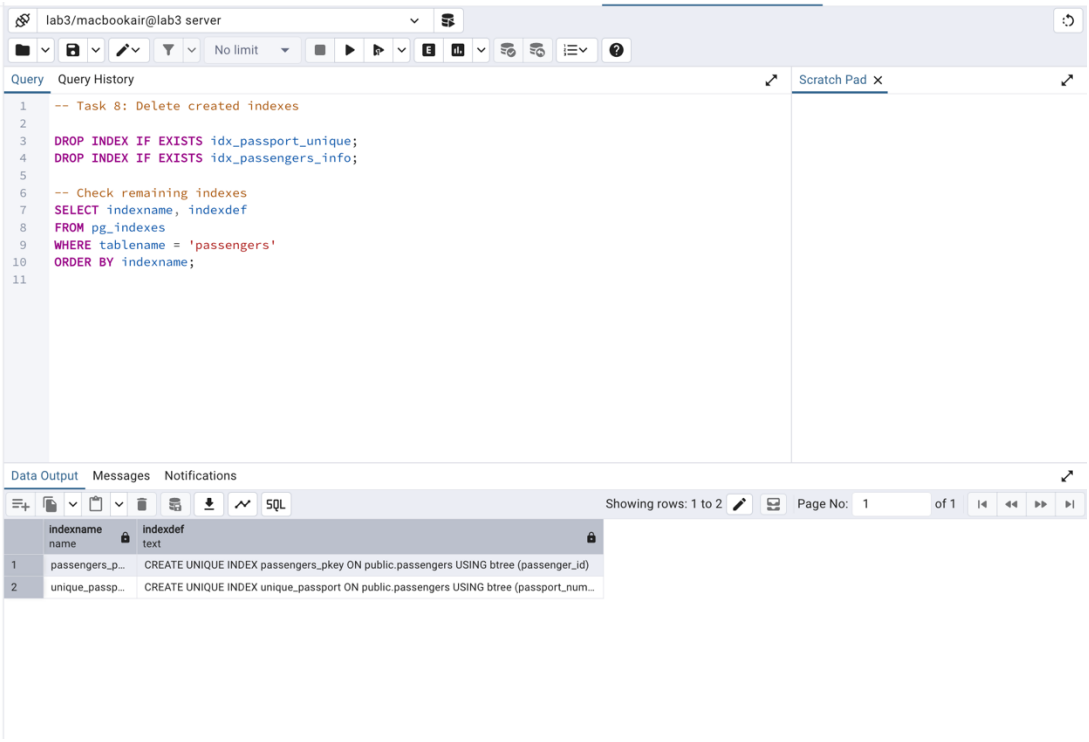

lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 6: Create unique index
CREATE UNIQUE INDEX IF NOT EXISTS idx_passport_unique
ON passengers (passport_number);
SELECT indexname, indexdef
FROM pg_indexes
WHERE tablename = 'passengers'
AND indexname = 'idx_passport_unique';
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 1
Page No: 1
of 1
indexname
name
indexdef
text
1
idx_passport_uniq...
CREATE UNIQUE INDEX idx_passport_unique ON public.passengers USING btree (passport_num...

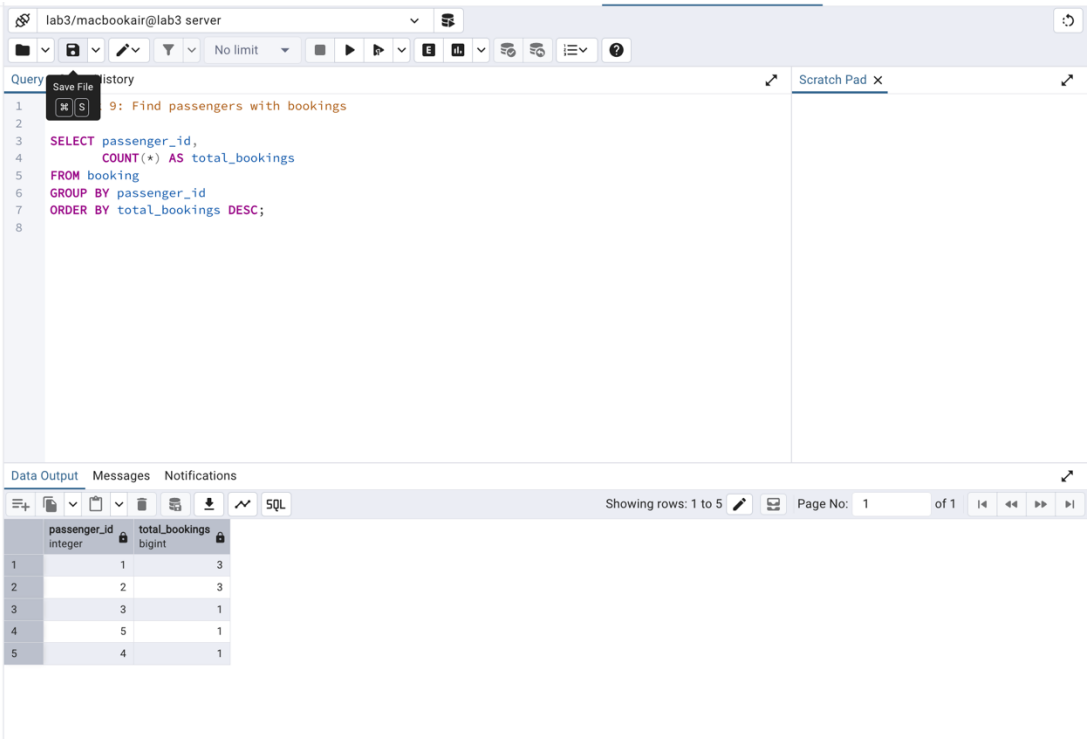

lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 6: Insert two passengers
WITH base AS (
SELECT COALESCE(MAX(passenger_id), 0) AS mx
FROM passengers
)
INSERT INTO passengers (
passenger_id,
first_name,
last_name,
date_of_birth,
gender,
country_of_citizenship,
country_of_residence,
passport_number,
created_at,
updated_at
)
SELECT
base.mx + n.id,
'Lab7',
CASE WHEN n.id = 1
THEN 'SampleOne'
ELSE 'SampleTwo'
END,
CASE WHEN n.id = 1
THEN DATE '1984-06-15'
ELSE DATE '1986-08-20'
END,
CASE WHEN n.id = 1
THEN 'Male'
ELSE 'Female'
END,
CASE WHEN n.id = 1
THEN 'Philippines'
ELSE 'Kazakhstan'
END,
Data Output
Messages
Notifications
Showing rows: 1 to 2
Page No: 1
of 1
passenger_id [PK] integer
first_name character varying (50)
last_name character varying (50)
passport_number character varying (20)
1 8 Lab7 SampleOne L7-8
2 9 Lab7 SampleTwo L7-9
Total rows: 2
Query complete 00:00:00.069
LF
Ln 1, Col 1


lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
gender,
country_of_citizenship,
country_of_residence,
passport_number,
created_at,
updated_at
)
SELECT
base.mx + n.id,
'Lab7',
CASE WHEN n.id = 1
THEN 'SampleOne'
ELSE 'SampleTwo'
END,
CASE WHEN n.id = 1
THEN DATE '1984-06-15'
ELSE DATE '1986-08-20'
END,
CASE WHEN n.id = 1
THEN 'Male'
ELSE 'Female'
END,
CASE WHEN n.id = 1
THEN 'Philippines'
ELSE 'Kazakhstan'
END,
CASE WHEN n.id = 1
THEN 'Philippines'
ELSE 'Kazakhstan'
END,
'L7-' || (base.mx + n.id)::text,
NOW(),
NOW()
FROM base
CROSS JOIN generate_series(1, 2) AS n(id)
RETURNING passenger_id, first_name,
last_name, passport_number;
Data Output
Messages
Notifications
Showing rows: 1 to 2
Page No: 1
of 1
passenger_id [PK] integer
first_name character varying (50)
last_name character varying (50)
passport_number character varying (20)
1 8 Lab7 SampleOne L7-8
2 9 Lab7 SampleTwo L7-9
Total rows: 2
Query complete 00:00:00.069
LF
Ln 1, Col 1


lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 7: Create composite index
CREATE INDEX IF NOT EXISTS idx_passengers_info
ON passengers (
country_of_citizenship,
date_of_birth,
first_name,
last_name
);
ANALYZE passengers;
EXPLAIN (ANALYZE, BUFFERS)
SELECT *
FROM passengers
WHERE country_of_citizenship = 'Philippines'
AND date_of_birth >= '1984-01-01'
AND date_of_birth < '1985-01-01';
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 8
Page No: 1
of 1
QUERY PLAN
text
1 Seq Scan on passengers (cost=0.00..1.16 rows=1 width=69) (actual time=0.016..0.017 rows=2.00 loops=1)
2 Filter: ((date_of_birth >= '1984-01-01'::date) AND (date_of_birth < '1985-01-01'::date) AND ((country_of_citizenship)::text = 'Philippines'::...
3 Rows Removed by Filter: 7
4 Buffers: shared hit=1
5 Planning:
6 Buffers: shared hit=56 read=1
7 Planning Time: 1.052 ms
8 Execution Time: 0.048 ms


lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 8: List indexes
SELECT indexname, indexdef
FROM pg_indexes
WHERE tablename = 'passengers'
ORDER BY indexname;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 4
Page No: 1
of 1
indexname
name
indexdef
text
1 idx_passengers_in... CREATE INDEX idx_passengers_info ON public.passengers USING btree (country_of_citizenship, date_of_birth, first_name, last_n...
2 idx_passport_uniq... CREATE UNIQUE INDEX idx_passport_unique ON public.passengers USING btree (passport_number)
3 passengers_pkey CREATE UNIQUE INDEX passengers_pkey ON public.passengers USING btree (passenger_id)
4 unique_passport CREATE UNIQUE INDEX unique_passport ON public.passengers USING btree (passport_number)


lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 8: Delete created indexes
DROP INDEX IF EXISTS idx_passport_unique;
DROP INDEX IF EXISTS idx_passengers_info;
-- Check remaining indexes
SELECT indexname, indexdef
FROM pg_indexes
WHERE tablename = 'passengers'
ORDER BY indexname;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 2
Page No: 1
of 1
indexname name
indexdef text
1 passengers_p... CREATE UNIQUE INDEX passengers_pkey ON public.passengers USING btree (passenger_id)
2 unique_passp... CREATE UNIQUE INDEX unique_passport ON public.passengers USING btree (passport_num...


lab3/macbookair@lab3 server
No limit
Save File
Scratch Pad
9: Find passengers with bookings
SELECT passenger_id,
COUNT(*) AS total_bookings
FROM booking
GROUP BY passenger_id
ORDER BY total_bookings DESC;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 5
Page No: 1
of 1
passenger_id integer
total_bookings bigint
1 1 3
2 2 3
3 3 1
4 5 1
5 4 1

lab3/macbookair@lab3 server

No limit

Query | Query History

Scratch Pad

```sql
-- Task 9: BEFORE (without index)

DROP INDEX IF EXISTS idx_booking_passenger;

ANALYZE booking;

EXPLAIN (ANALYZE, BUFFERS)
SELECT *
FROM booking
WHERE passenger_id = 1;
```

Data Output | Messages | Notifications

Showing rows: 1 to 8 | Page No: 1 of 1

	QUERY PLAN text
1	Seq Scan on booking (cost=0.00..1.11 rows=3 width=57) (actual time=0.012..0.014 rows=3.00 loop…
2	Filter: (passenger_id = 1)
3	Rows Removed by Filter: 6
4	Buffers: shared hit=1
5	Planning:
6	Buffers: shared hit=34
7	Planning Time: 0.117 ms
8	Execution Time: 0.031 ms

lab3/macbookair@lab3 server

No limit

Save File ⌘ S

Query | History

Scratch Pad

```sql
-- Task 9: AFTER (with index)

CREATE INDEX IF NOT EXISTS idx_booking_passenger
ON booking (passenger_id);

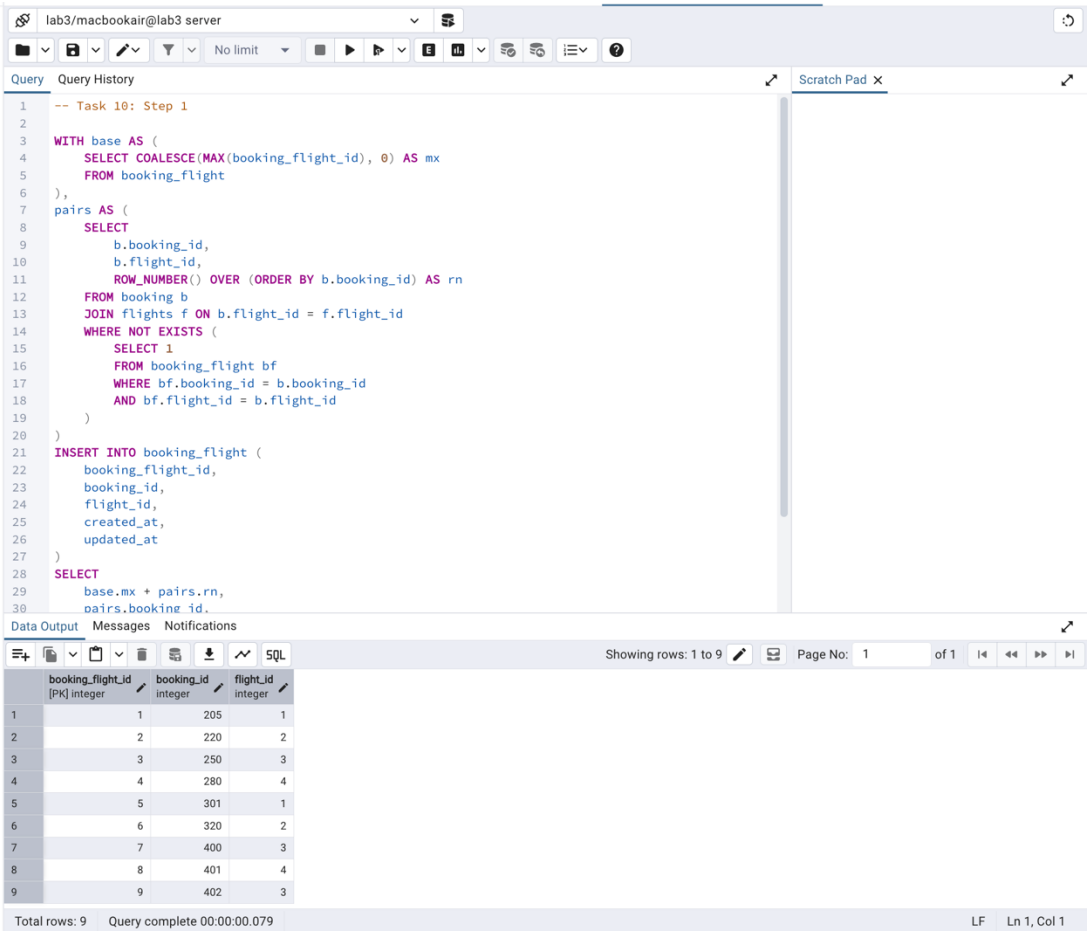
ANALYZE booking;

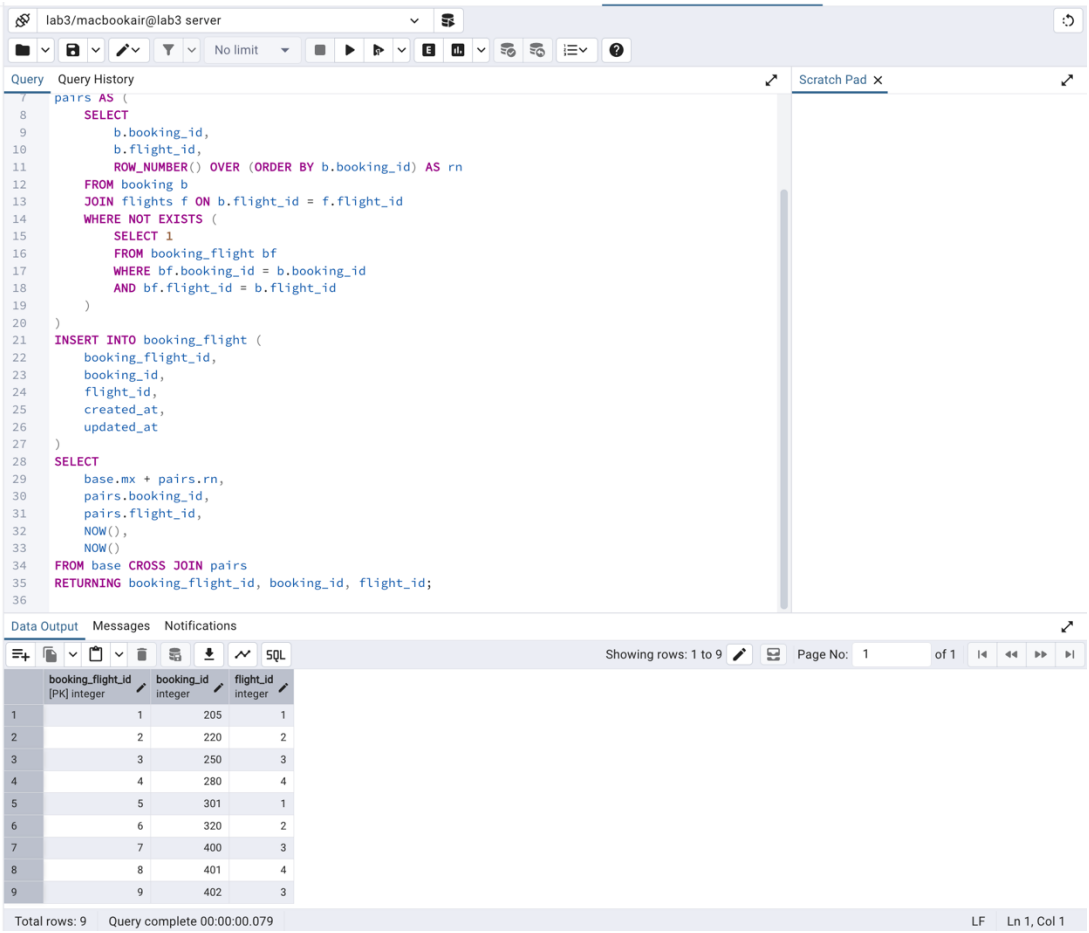
EXPLAIN (ANALYZE, BUFFERS)
SELECT *
FROM booking
WHERE passenger_id = 1;
```

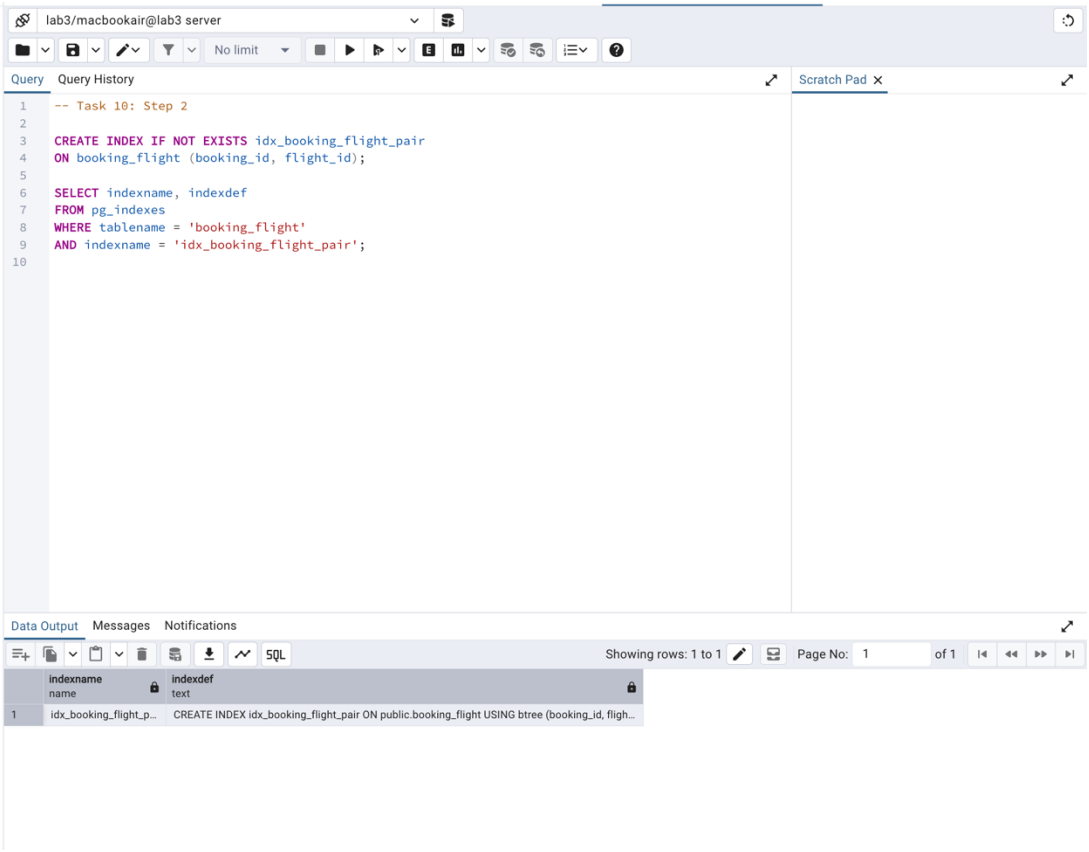
Data Output | Messages | Notifications

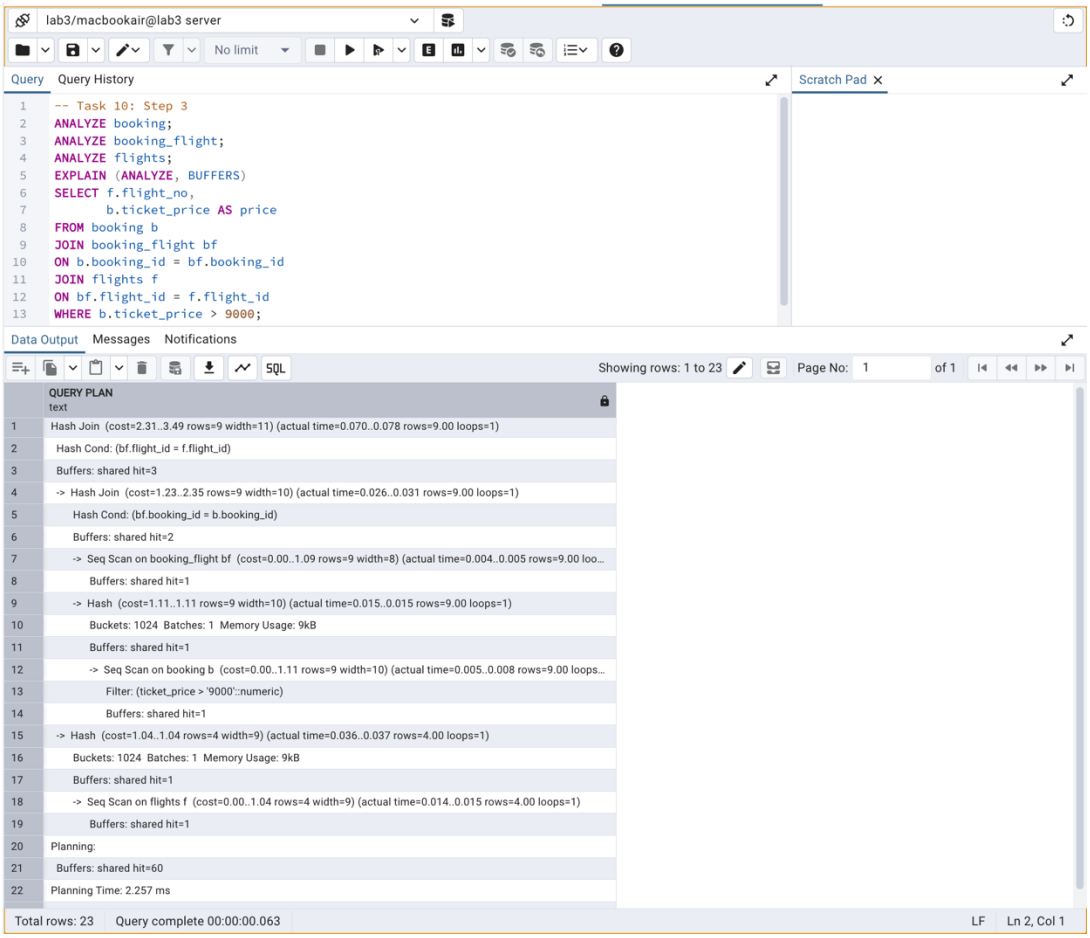
Showing rows: 1 to 8 | Page No: 1 of 1

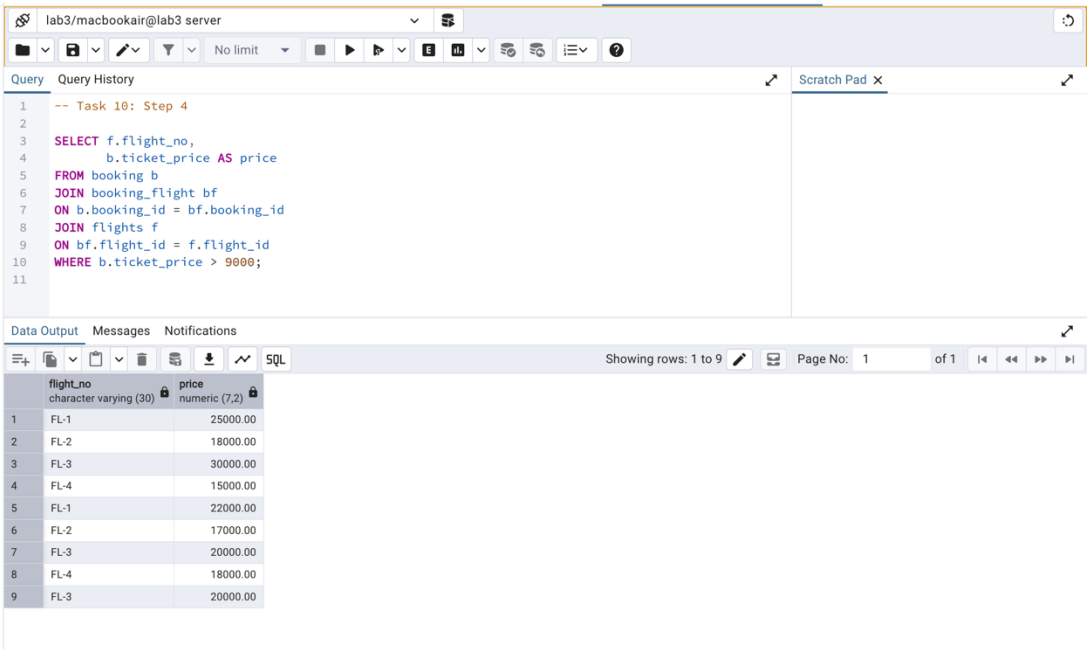
	QUERY PLAN text
1	Seq Scan on booking (cost=0.00..1.11 rows=3 width=57) (actual time=0.010..0.012 rows=3.00 loop…
2	Filter: (passenger_id = 1)
3	Rows Removed by Filter: 6
4	Buffers: shared hit=1
5	Planning:
6	Buffers: shared hit=29 read=1
7	Planning Time: 0.228 ms
8	Execution Time: 0.028 ms

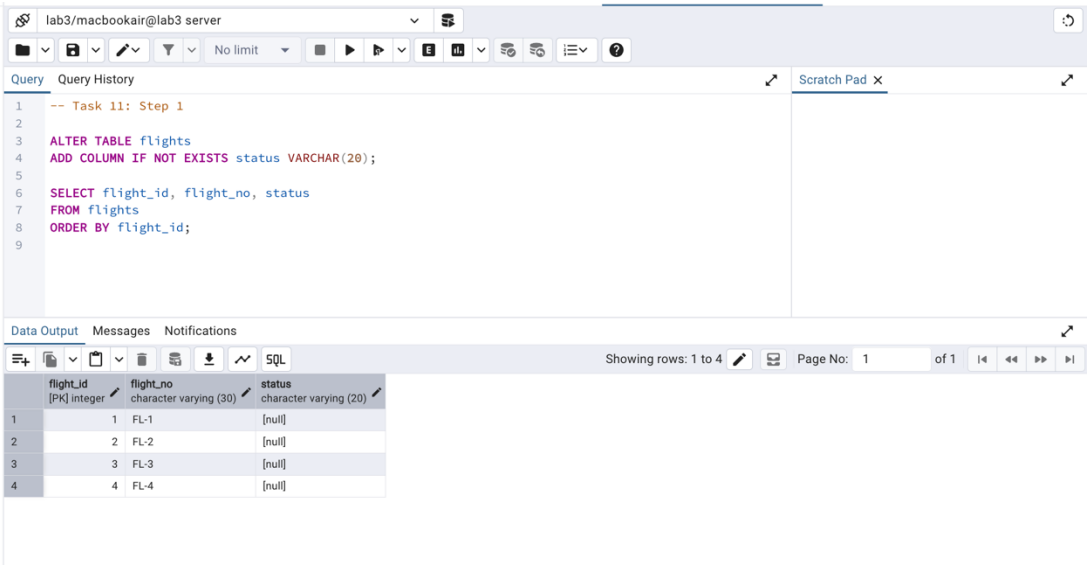

lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 10: Step 1
WITH base AS (
SELECT COALESCE(MAX(booking_flight_id), 0) AS mx
FROM booking_flight
),
pairs AS (
SELECT
b.booking_id,
b.flight_id,
ROW_NUMBER() OVER (ORDER BY b.booking_id) AS rn
FROM booking b
JOIN flights f ON b.flight_id = f.flight_id
WHERE NOT EXISTS (
SELECT 1
FROM booking_flight bf
WHERE bf.booking_id = b.booking_id
AND bf.flight_id = b.flight_id
)
)
INSERT INTO booking_flight (
booking_flight_id,
booking_id,
flight_id,
created_at,
updated_at
)
SELECT
base.mx + pairs.rn,
pairs.booking_id,
Data Output
Messages
Notifications
Showing rows: 1 to 9
Page No: 1
of 1
booking_flight_id [PK] integer
booking_id integer
flight_id integer
1 1 205 1
2 2 220 2
3 3 250 3
4 4 280 4
5 5 301 1
6 6 320 2
7 7 400 3
8 8 401 4
9 9 402 3
Total rows: 9
Query complete 00:00:00.079
LF
Ln 1, Col 1

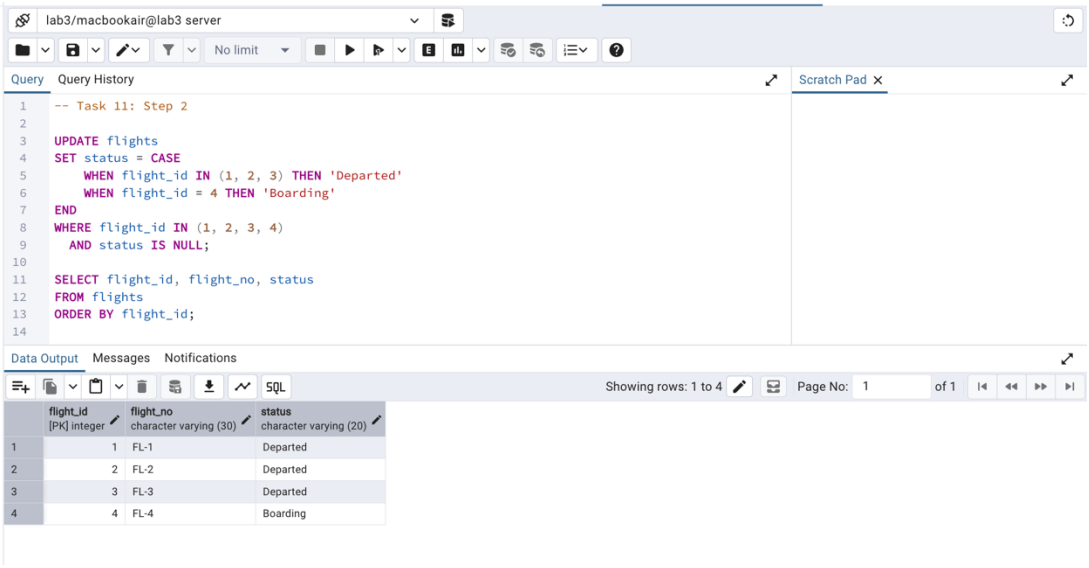

lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
pairs AS (
SELECT
b.booking_id,
b.flight_id,
ROW_NUMBER() OVER (ORDER BY b.booking_id) AS rn
FROM booking b
JOIN flights f ON b.flight_id = f.flight_id
WHERE NOT EXISTS (
SELECT 1
FROM booking_flight bf
WHERE bf.booking_id = b.booking_id
AND bf.flight_id = b.flight_id
)
)
INSERT INTO booking_flight (
booking_flight_id,
booking_id,
flight_id,
created_at,
updated_at
)
SELECT
base.mx + pairs.rn,
pairs.booking_id,
pairs.flight_id,
NOW(),
NOW()
FROM base CROSS JOIN pairs
RETURNING booking_flight_id, booking_id, flight_id;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 9
Page No: 1
of 1
booking_flight_id [PK] integer
booking_id integer
flight_id integer
1 1 205 1
2 2 220 2
3 3 250 3
4 4 280 4
5 5 301 1
6 6 320 2
7 7 400 3
8 8 401 4
9 9 402 3
Total rows: 9
Query complete 00:00:00.079
LF
Ln 1, Col 1

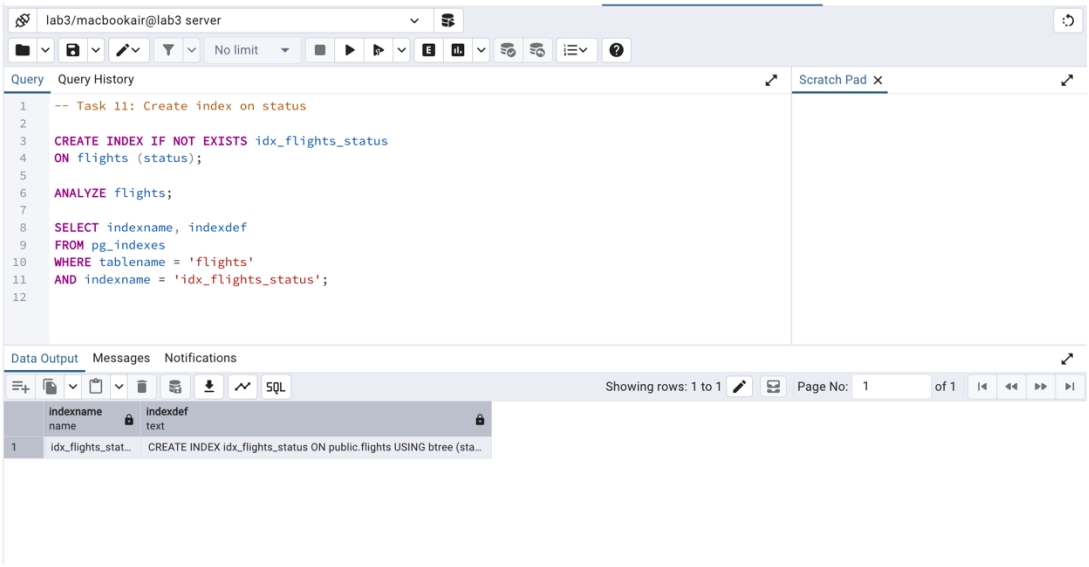

lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 10: Step 2
CREATE INDEX IF NOT EXISTS idx_booking_flight_pair
ON booking_flight (booking_id, flight_id);
SELECT indexname, indexdef
FROM pg_indexes
WHERE tablename = 'booking_flight'
AND indexname = 'idx_booking_flight_pair';
Data Output
Messages
Notifications
Showing rows: 1 to 1
Page No: 1
of 1
indexname
name
indexdef
text
1
idx_booking_flight_p...
CREATE INDEX idx_booking_flight_pair ON public.booking_flight USING btree (booking_id, fligh...

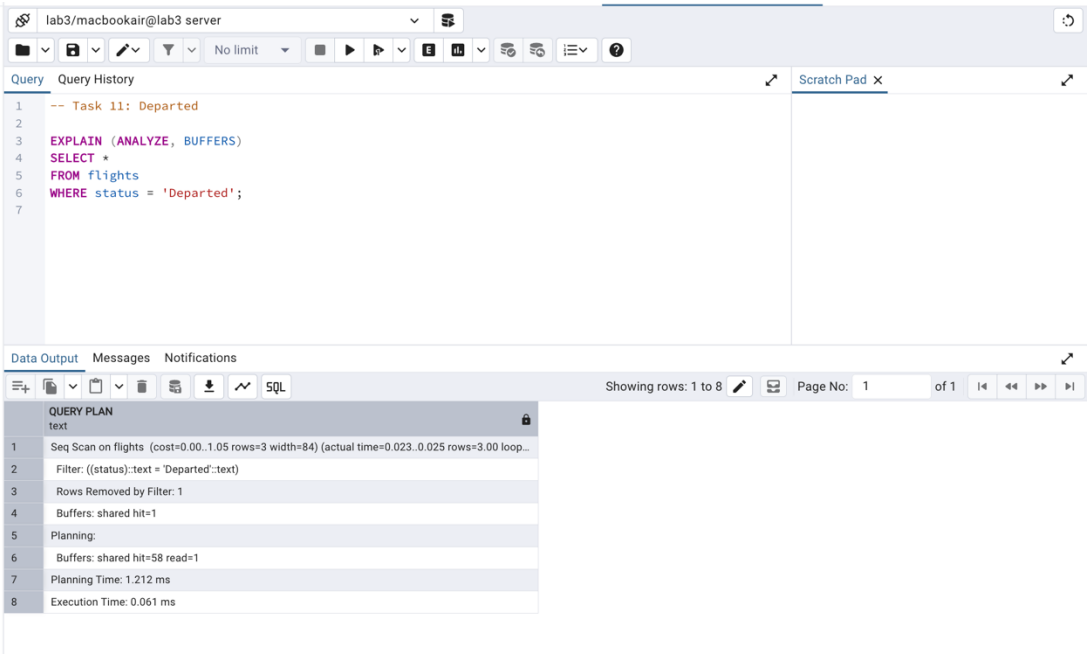

lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 10: Step 3
ANALYZE booking;
ANALYZE booking_flight;
ANALYZE flights;
EXPLAIN (ANALYZE, BUFFERS)
SELECT f.flight_no,
b.ticket_price AS price
FROM booking b
JOIN booking_flight bf
ON b.booking_id = bf.booking_id
JOIN flights f
ON bf.flight_id = f.flight_id
WHERE b.ticket_price > 9000;
Data Output
Messages
Notifications
Showing rows: 1 to 23
Page No: 1
of 1
QUERY PLAN
text
Hash Join (cost=2.31..3.49 rows=9 width=11) (actual time=0.070..0.078 rows=9.00 loops=1)
Hash Cond: (bf.flight_id = f.flight_id)
Buffers: shared hit=3
-> Hash Join (cost=1.23..2.35 rows=9 width=10) (actual time=0.026..0.031 rows=9.00 loops=1)
Hash Cond: (bf.booking_id = b.booking_id)
Buffers: shared hit=2
-> Seq Scan on booking_flight bf (cost=0.00..1.09 rows=9 width=8) (actual time=0.004..0.005 rows=9.00 loo...
Buffers: shared hit=1
-> Hash (cost=1.11..1.11 rows=9 width=10) (actual time=0.015..0.015 rows=9.00 loops=1)
Buckets: 1024 Batches: 1 Memory Usage: 9kB
Buffers: shared hit=1
-> Seq Scan on booking b (cost=0.00..1.11 rows=9 width=10) (actual time=0.005..0.008 rows=9.00 loops...
Filter: (ticket_price > '9000'::numeric)
Buffers: shared hit=1
-> Hash (cost=1.04..1.04 rows=4 width=9) (actual time=0.036..0.037 rows=4.00 loops=1)
Buckets: 1024 Batches: 1 Memory Usage: 9kB
Buffers: shared hit=1
-> Seq Scan on flights f (cost=0.00..1.04 rows=4 width=9) (actual time=0.014..0.015 rows=4.00 loops=1)
Buffers: shared hit=1
Planning:
Buffers: shared hit=60
Planning Time: 2.257 ms
Total rows: 23
Query complete 00:00:00.063
LF
Ln 2, Col 1

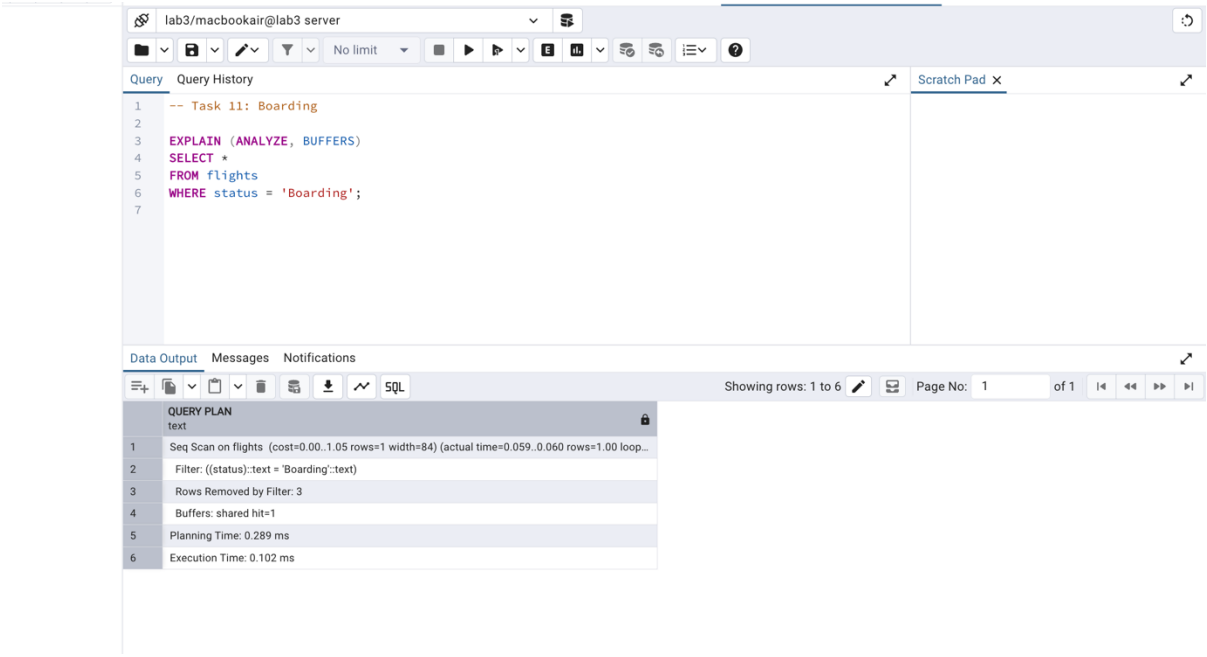

lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 10: Step 4
SELECT f.flight_no,
b.ticket_price AS price
FROM booking b
JOIN booking_flight bf
ON b.booking_id = bf.booking_id
JOIN flights f
ON bf.flight_id = f.flight_id
WHERE b.ticket_price > 9000;
Data Output
Messages
Notifications
Showing rows: 1 to 9
Page No: 1
of 1
flight_no
character varying (30)
price
numeric (7,2)
FL-1 25000.00
FL-2 18000.00
FL-3 30000.00
FL-4 15000.00
FL-1 22000.00
FL-2 17000.00
FL-3 20000.00
FL-4 18000.00
FL-3 20000.00

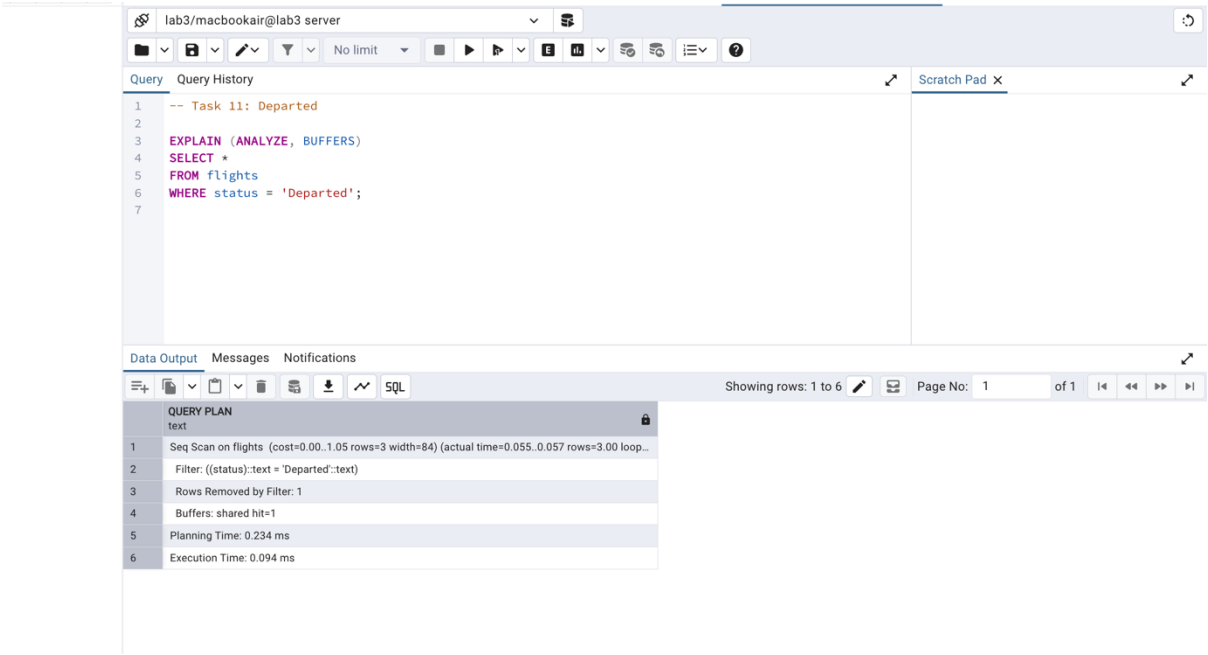

lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 11: Step 1
ALTER TABLE flights
ADD COLUMN IF NOT EXISTS status VARCHAR(20);
SELECT flight_id, flight_no, status
FROM flights
ORDER BY flight_id;
Data Output
Messages
Notifications
Showing rows: 1 to 4
Page No: 1
of 1
flight_id [PK] integer
flight_no character varying (30)
status character varying (20)
1 1 FL-1 [null]
2 2 FL-2 [null]
3 3 FL-3 [null]
4 4 FL-4 [null]

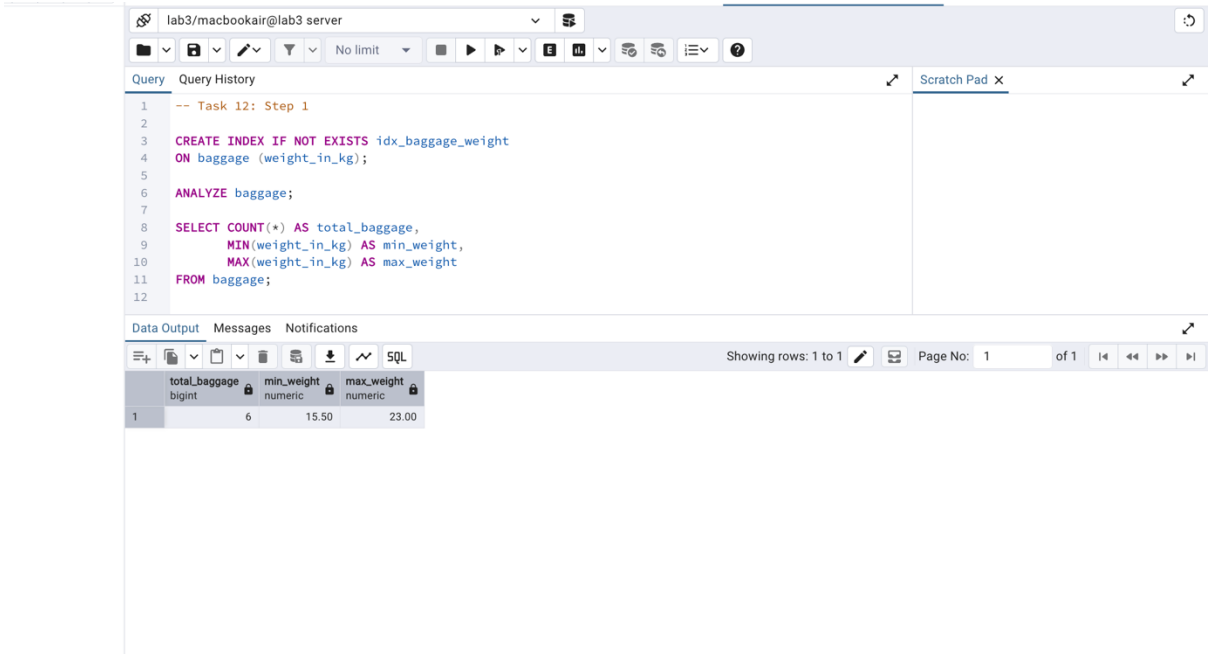

lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 11: Step 2
UPDATE flights
SET status = CASE
WHEN flight_id IN (1, 2, 3) THEN 'Departed'
WHEN flight_id = 4 THEN 'Boarding'
END
WHERE flight_id IN (1, 2, 3, 4)
AND status IS NULL;
SELECT flight_id, flight_no, status
FROM flights
ORDER BY flight_id;
Data Output
Messages
Notifications
Showing rows: 1 to 4
Page No: 1
of 1
flight_id [PK] integer
flight_no character varying (30)
status character varying (20)
1 1 FL-1 Departed
2 2 FL-2 Departed
3 3 FL-3 Departed
4 4 FL-4 Boarding

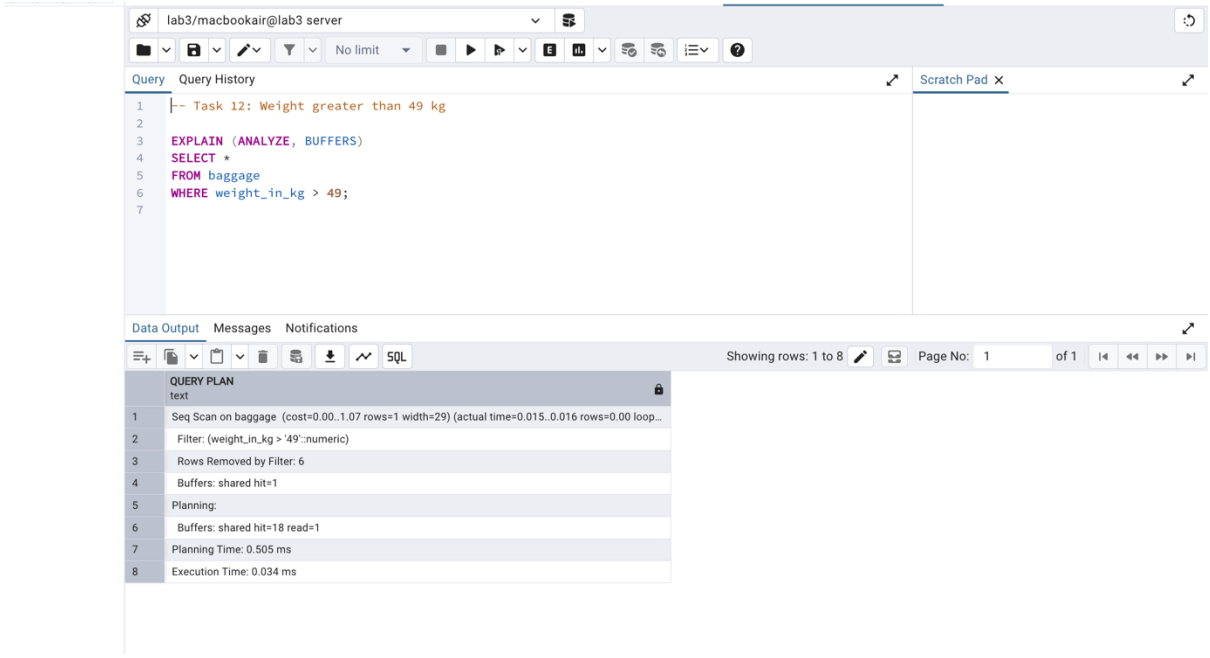

lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 11: Create index on status
CREATE INDEX IF NOT EXISTS idx_flights_status
ON flights (status);
ANALYZE flights;
SELECT indexname, indexdef
FROM pg_indexes
WHERE tablename = 'flights'
AND indexname = 'idx_flights_status';
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 1
Page No: 1
of 1
indexname
name
indexdef
text
1
idx_flights_stat...
CREATE INDEX idx_flights_status ON public.flights USING btree (sta...

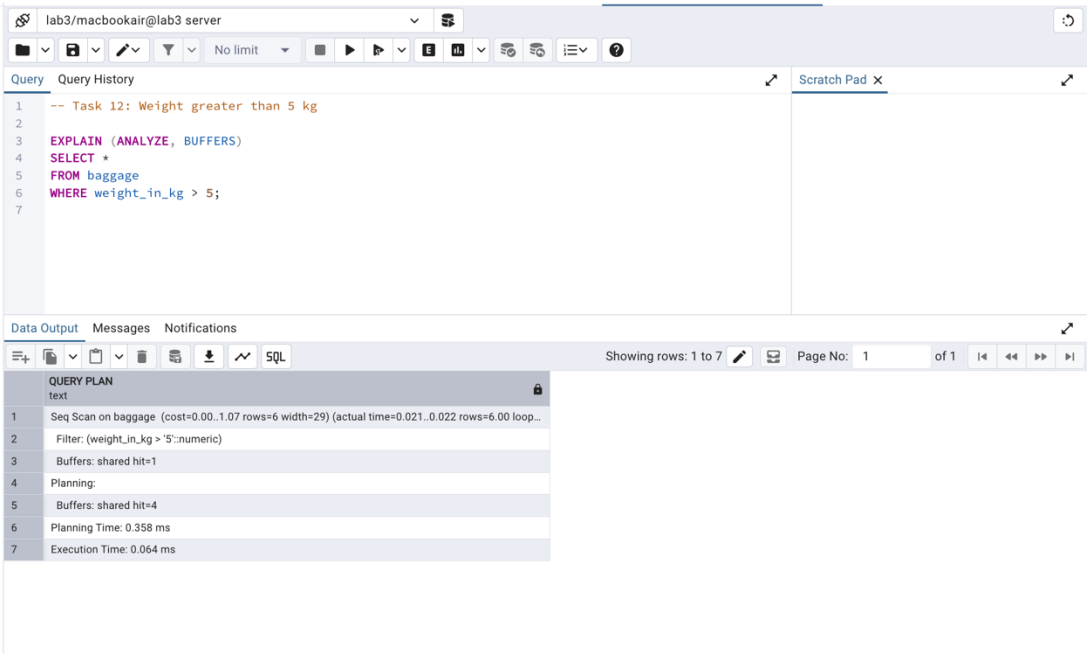

lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 11: Departed
EXPLAIN (ANALYZE, BUFFERS)
SELECT *
FROM flights
WHERE status = 'Departed';
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 8
Page No: 1
of 1
QUERY PLAN
text
1 Seq Scan on flights (cost=0.00..1.05 rows=3 width=84) (actual time=0.023..0.025 rows=3.00 loop...
2 Filter: ((status)::text = 'Departed'::text)
3 Rows Removed by Filter: 1
4 Buffers: shared hit=1
5 Planning:
6 Buffers: shared hit=58 read=1
7 Planning Time: 1.212 ms
8 Execution Time: 0.061 ms

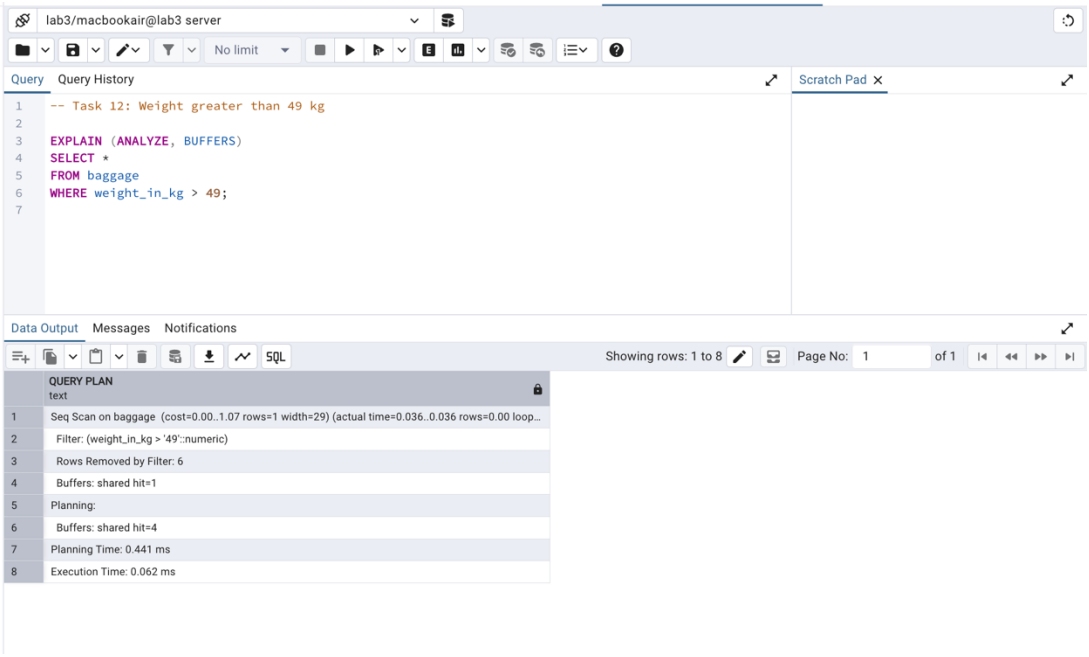

lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 11: Boarding
EXPLAIN (ANALYZE, BUFFERS)
SELECT *
FROM flights
WHERE status = 'Boarding';
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 6
Page No: 1
of 1
QUERY PLAN
text
1 Seq Scan on flights (cost=0.00..1.05 rows=1 width=84) (actual time=0.059..0.060 rows=1.00 loop...
2 Filter: ((status)::text = 'Boarding'::text)
3 Rows Removed by Filter: 3
4 Buffers: shared hit=1
5 Planning Time: 0.289 ms
6 Execution Time: 0.102 ms

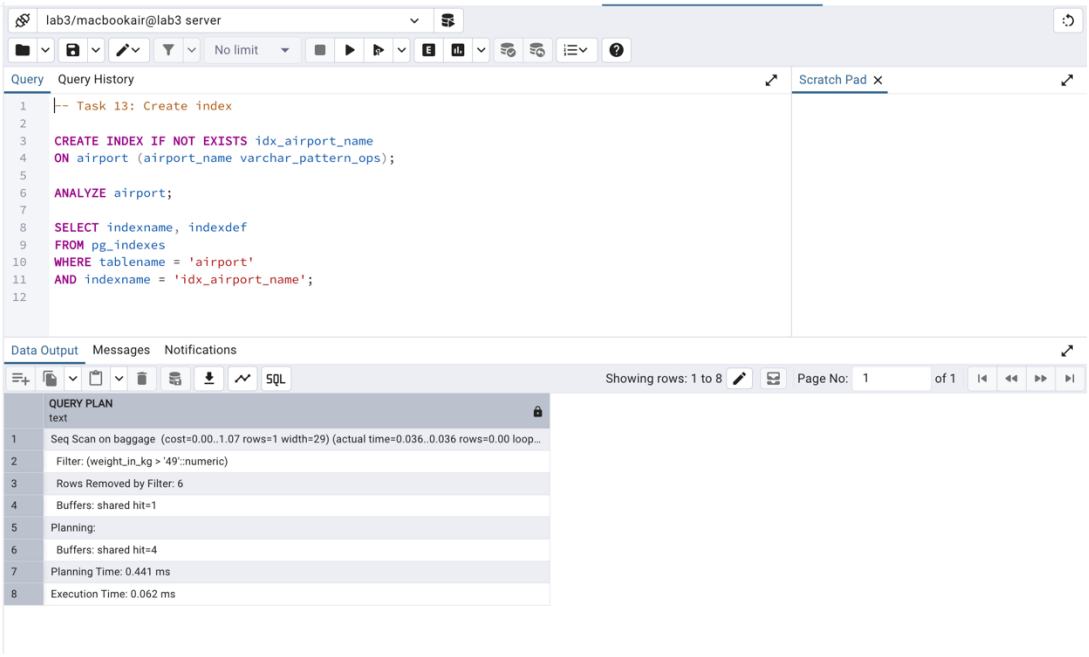

lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 11: Departed
EXPLAIN (ANALYZE, BUFFERS)
SELECT *
FROM flights
WHERE status = 'Departed';
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 6
Page No: 1
of 1
QUERY PLAN
text
1 Seq Scan on flights (cost=0.00..1.05 rows=3 width=84) (actual time=0.055..0.057 rows=3.00 loop...
2 Filter: ((status)::text = 'Departed'::text)
3 Rows Removed by Filter: 1
4 Buffers: shared hit=1
5 Planning Time: 0.234 ms
6 Execution Time: 0.094 ms

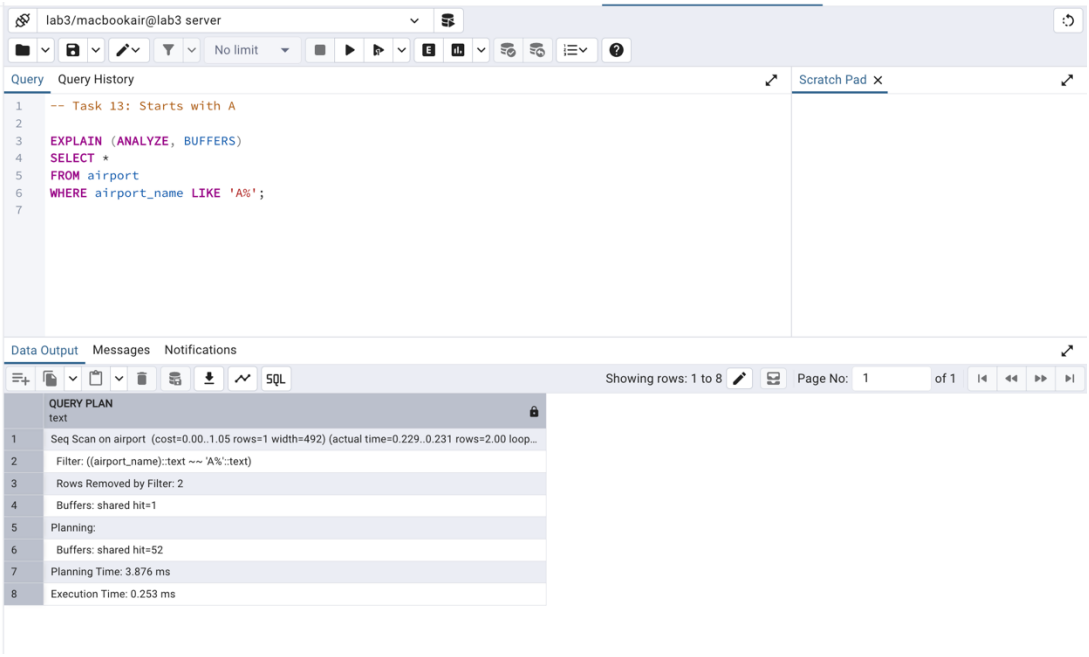

lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 12: Step 1
CREATE INDEX IF NOT EXISTS idx_baggage_weight
ON baggage (weight_in_kg);
ANALYZE baggage;
SELECT COUNT(*) AS total_baggage,
MIN(weight_in_kg) AS min_weight,
MAX(weight_in_kg) AS max_weight
FROM baggage;
Data Output
Messages
Notifications
Showing rows: 1 to 1
Page No: 1
of 1
total_baggage bigint
min_weight numeric
max_weight numeric
1
6
15.50
23.00

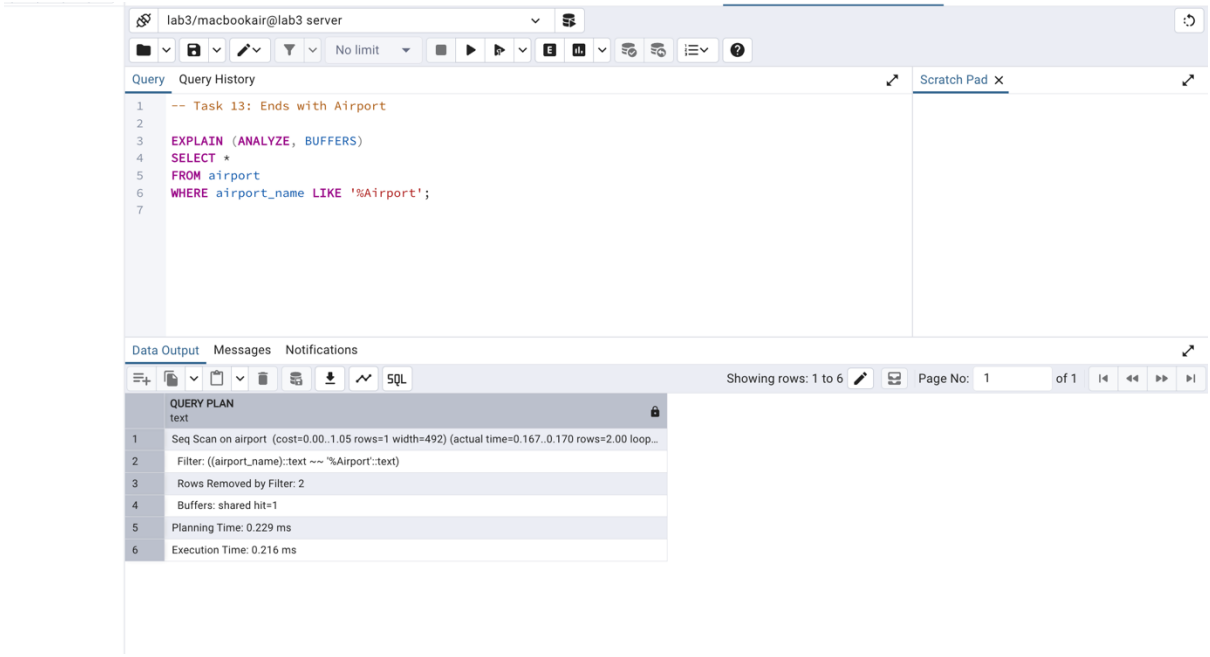

lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 12: Weight greater than 49 kg
EXPLAIN (ANALYZE, BUFFERS)
SELECT *
FROM baggage
WHERE weight_in_kg > 49;
Data Output
Messages
Notifications
Showing rows: 1 to 8
Page No: 1
of 1
QUERY PLAN text
1 Seq Scan on baggage (cost=0.00..1.07 rows=1 width=29) (actual time=0.015..0.016 rows=0.00 loop...
2 Filter: (weight_in_kg > '49'::numeric)
3 Rows Removed by Filter: 6
4 Buffers: shared hit=1
5 Planning:
6 Buffers: shared hit=18 read=1
7 Planning Time: 0.505 ms
8 Execution Time: 0.034 ms

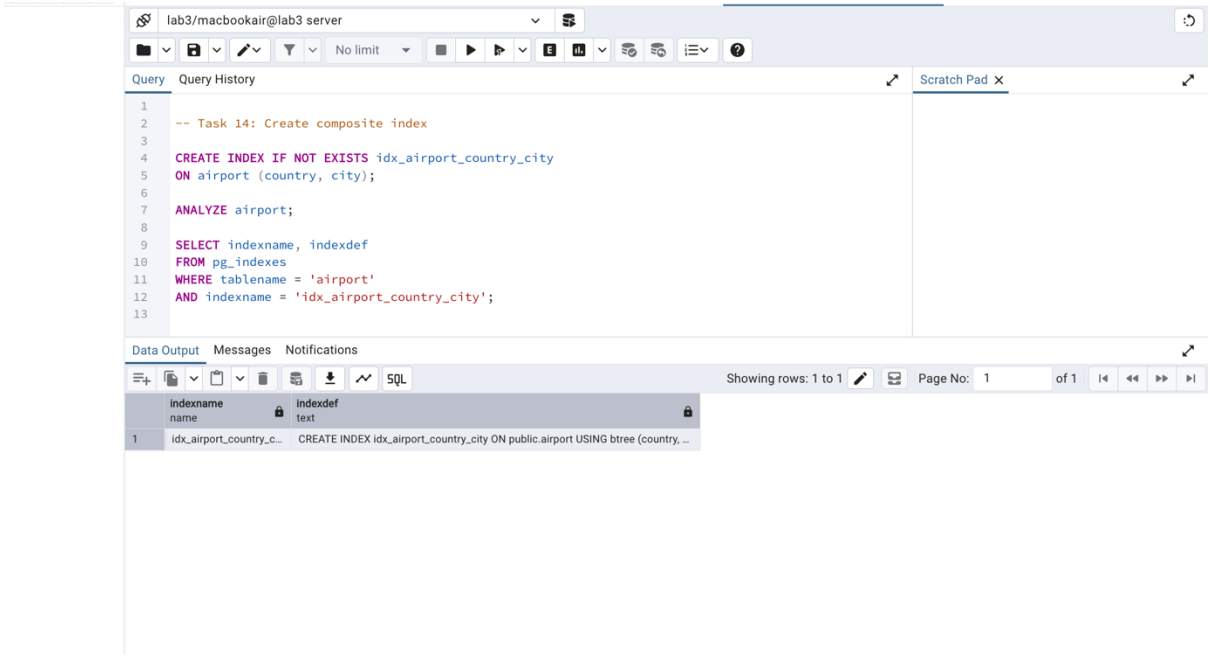

lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 12: Weight greater than 5 kg
EXPLAIN (ANALYZE, BUFFERS)
SELECT *
FROM baggage
WHERE weight_in_kg > 5;
Data Output
Messages
Notifications
Showing rows: 1 to 7
Page No: 1
of 1
QUERY PLAN
text
1 Seq Scan on baggage (cost=0.00..1.07 rows=6 width=29) (actual time=0.021..0.022 rows=6.00 loop...
2 Filter: (weight_in_kg > '5'::numeric)
3 Buffers: shared hit=1
4 Planning:
5 Buffers: shared hit=4
6 Planning Time: 0.358 ms
7 Execution Time: 0.064 ms

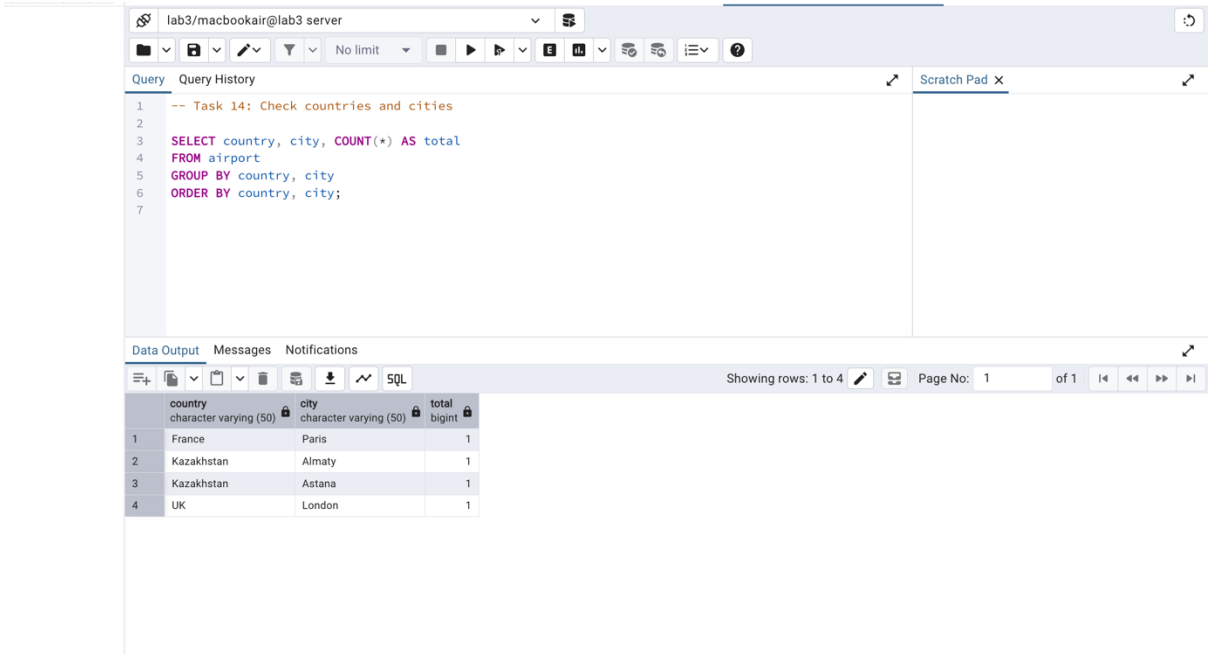

lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 12: Weight greater than 49 kg
EXPLAIN (ANALYZE, BUFFERS)
SELECT *
FROM baggage
WHERE weight_in_kg > 49;
Data Output
Messages
Notifications
Showing rows: 1 to 8
Page No: 1
of 1
QUERY PLAN
text
1 Seq Scan on baggage (cost=0.00..1.07 rows=1 width=29) (actual time=0.036..0.036 rows=0.00 loop...
2 Filter: (weight_in_kg > '49'::numeric)
3 Rows Removed by Filter: 6
4 Buffers: shared hit=1
5 Planning:
6 Buffers: shared hit=4
7 Planning Time: 0.441 ms
8 Execution Time: 0.062 ms

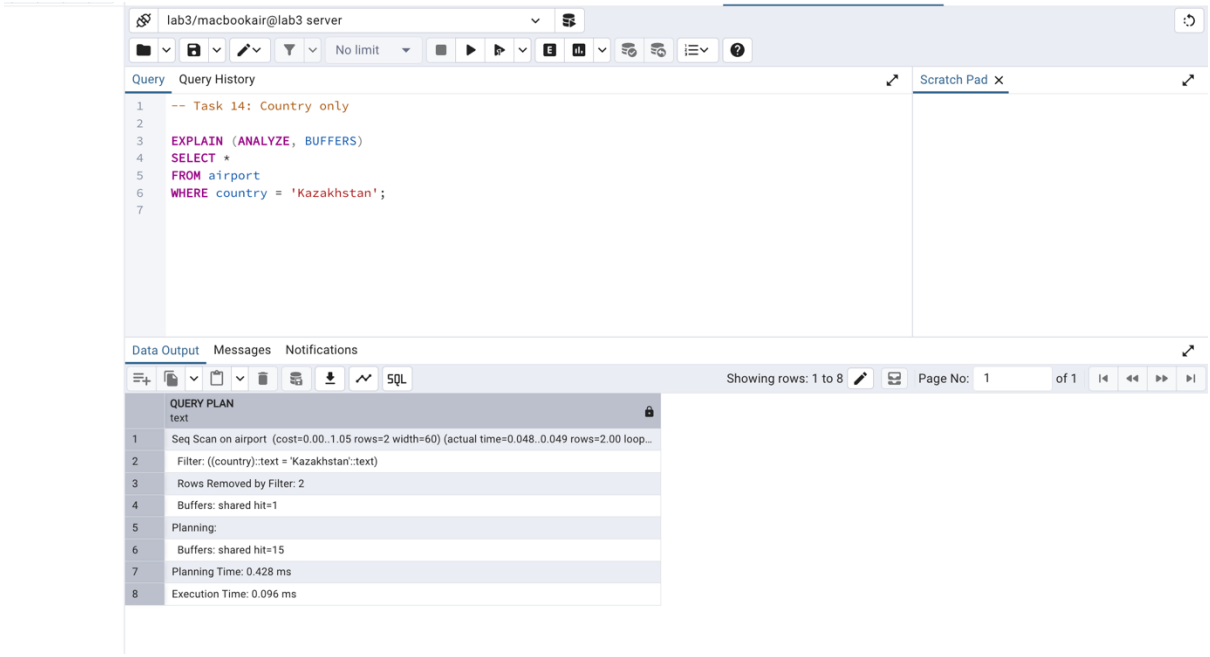

lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 13: Create index
CREATE INDEX IF NOT EXISTS idx_airport_name
ON airport (airport_name varchar_pattern_ops);
ANALYZE airport;
SELECT indexname, indexdef
FROM pg_indexes
WHERE tablename = 'airport'
AND indexname = 'idx_airport_name';
Data Output
Messages
Notifications
Showing rows: 1 to 8
Page No: 1
of 1
QUERY PLAN
text
1 Seq Scan on baggage (cost=0.00..1.07 rows=1 width=29) (actual time=0.036..0.036 rows=0.00 loop...
2 Filter: (weight_in_kg > '49'::numeric)
3 Rows Removed by Filter: 6
4 Buffers: shared hit=1
5 Planning:
6 Buffers: shared hit=4
7 Planning Time: 0.441 ms
8 Execution Time: 0.062 ms

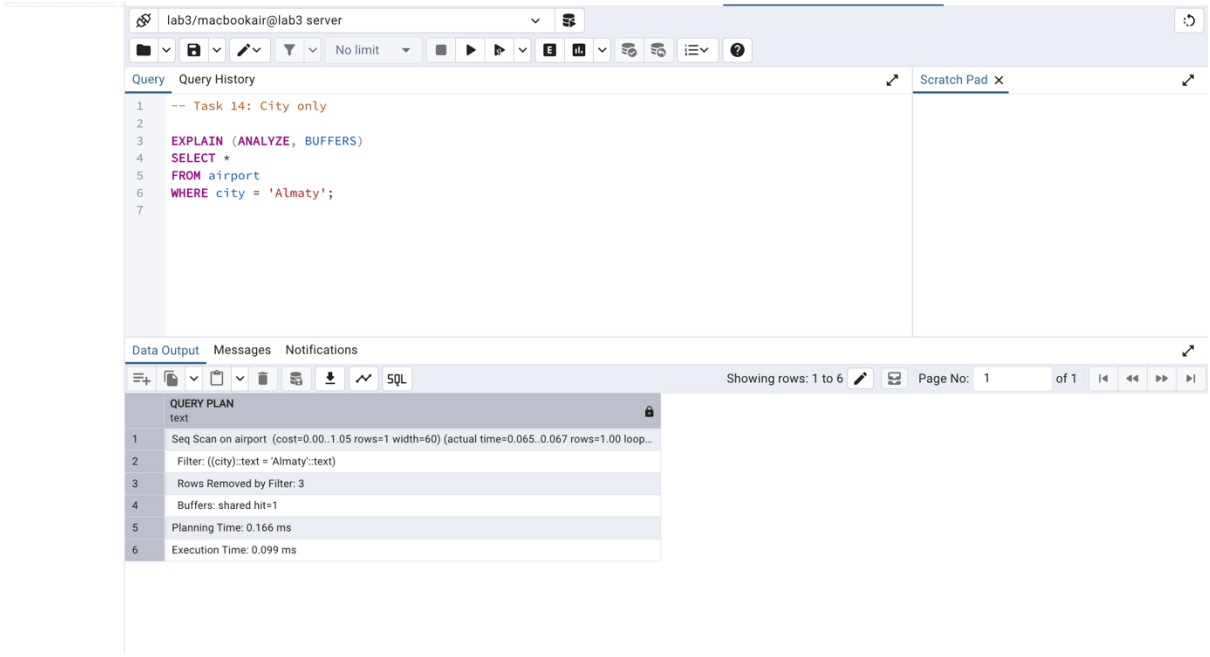

lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 13: Starts with A
EXPLAIN (ANALYZE, BUFFERS)
SELECT *
FROM airport
WHERE airport_name LIKE 'A%';
Data Output
Messages
Notifications
Showing rows: 1 to 8
Page No: 1
of 1
QUERY PLAN
text
1 Seq Scan on airport (cost=0.00..1.05 rows=1 width=492) (actual time=0.229..0.231 rows=2.00 loop...
2 Filter: ((airport_name)::text ~~ 'A%'::text)
3 Rows Removed by Filter: 2
4 Buffers: shared hit=1
5 Planning:
6 Buffers: shared hit=52
7 Planning Time: 3.876 ms
8 Execution Time: 0.253 ms

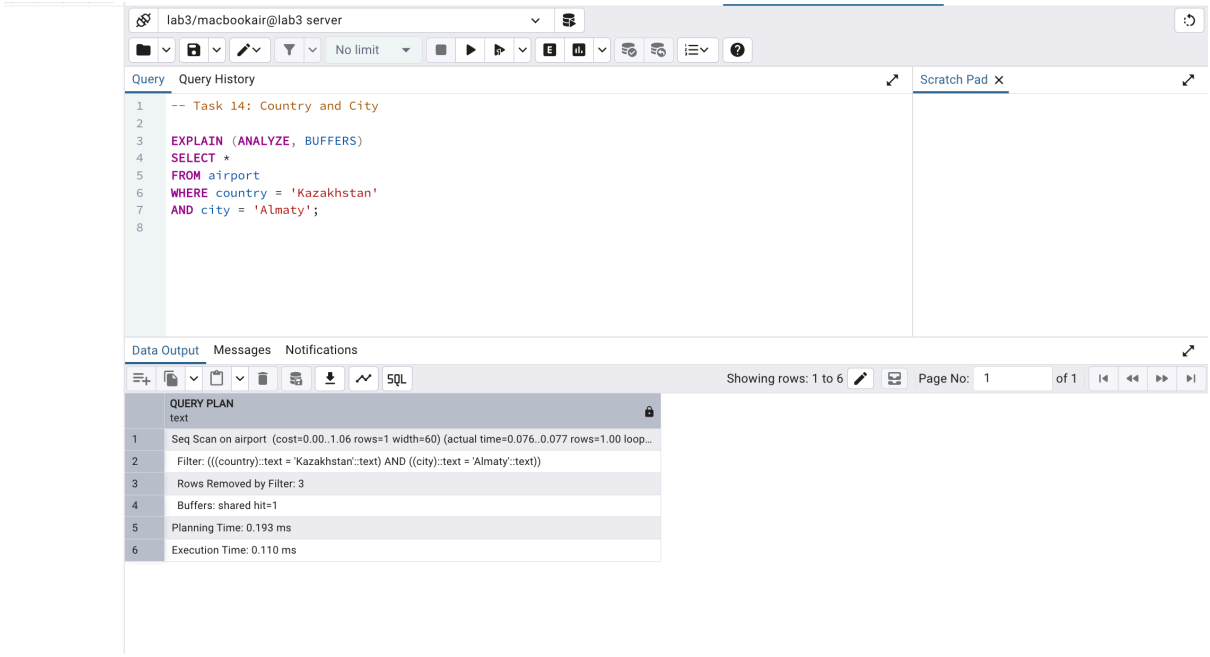

lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 13: Ends with Airport
EXPLAIN (ANALYZE, BUFFERS)
SELECT *
FROM airport
WHERE airport_name LIKE '%Airport';
Data Output
Messages
Notifications
Showing rows: 1 to 6
Page No: 1
of 1
QUERY PLAN
text
1 Seq Scan on airport (cost=0.00..1.05 rows=1 width=492) (actual time=0.167..0.170 rows=2.00 loop...
2 Filter: ((airport_name)::text ~~ '%Airport'::text)
3 Rows Removed by Filter: 2
4 Buffers: shared hit=1
5 Planning Time: 0.229 ms
6 Execution Time: 0.216 ms

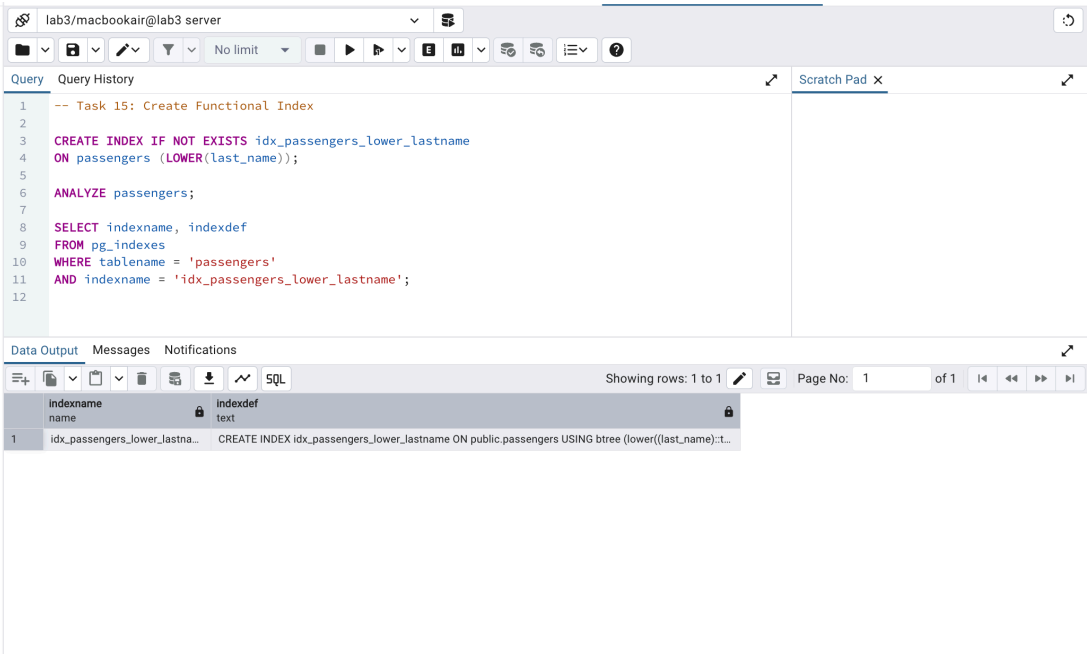

lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 14: Create composite index
CREATE INDEX IF NOT EXISTS idx_airport_country_city
ON airport (country, city);
ANALYZE airport;
SELECT indexname, indexdef
FROM pg_indexes
WHERE tablename = 'airport'
AND indexname = 'idx_airport_country_city';
Data Output
Messages
Notifications
Showing rows: 1 to 1
Page No: 1
of 1
indexname
name
indexdef
text
1 idx_airport_country_c... CREATE INDEX idx_airport_country_city ON public.airport USING btree (country, ...

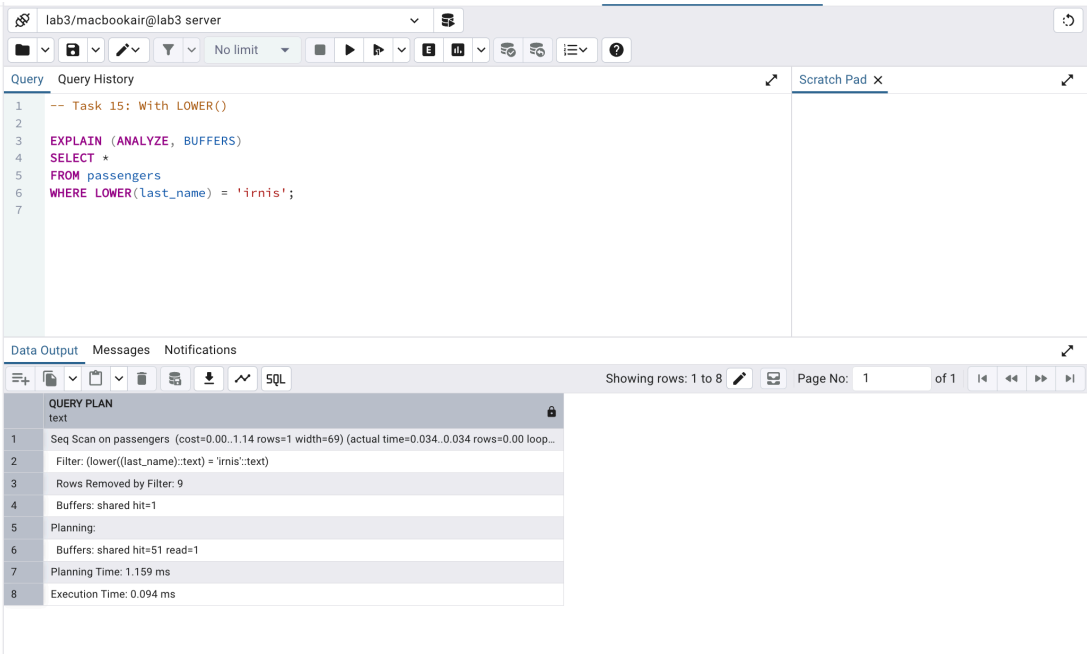

lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 14: Check countries and cities
SELECT country, city, COUNT(*) AS total
FROM airport
GROUP BY country, city
ORDER BY country, city;
Data Output
Messages
Notifications
Showing rows: 1 to 4
Page No: 1
of 1
country character varying (50)
city character varying (50)
total bigint
1 France Paris 1
2 Kazakhstan Almaty 1
3 Kazakhstan Astana 1
4 UK London 1

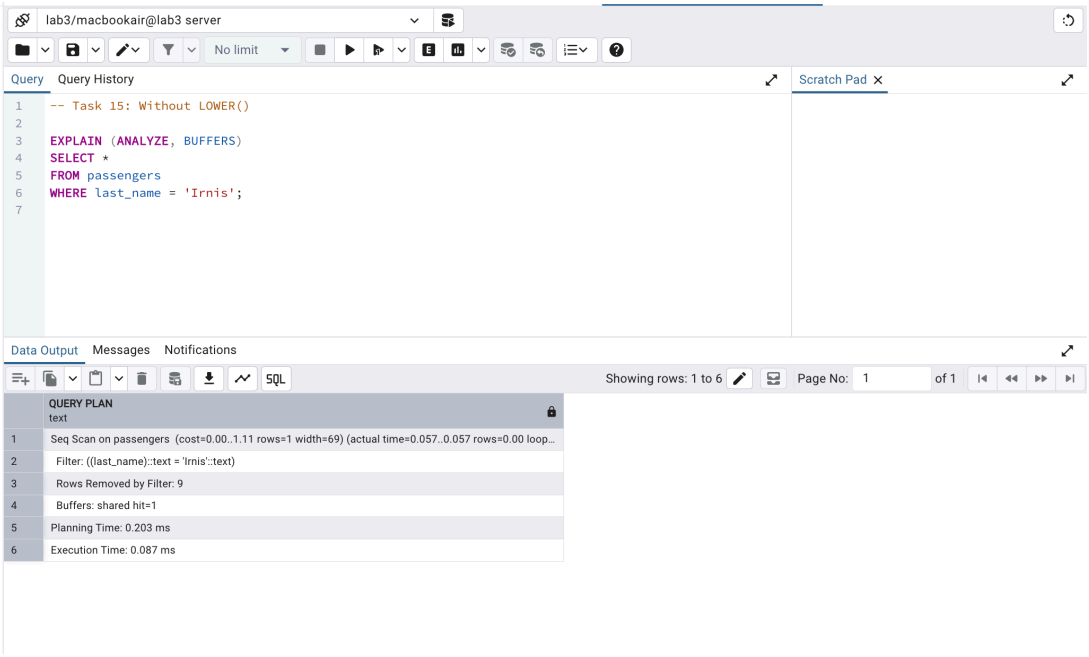

lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 14: Country only
EXPLAIN (ANALYZE, BUFFERS)
SELECT *
FROM airport
WHERE country = 'Kazakhstan';
Data Output
Messages
Notifications
Showing rows: 1 to 8
Page No: 1
of 1
QUERY PLAN text
1 Seq Scan on airport (cost=0.00..1.05 rows=2 width=60) (actual time=0.048..0.049 rows=2.00 loop...
2 Filter: ((country)::text = 'Kazakhstan'::text)
3 Rows Removed by Filter: 2
4 Buffers: shared hit=1
5 Planning:
6 Buffers: shared hit=15
7 Planning Time: 0.428 ms
8 Execution Time: 0.096 ms

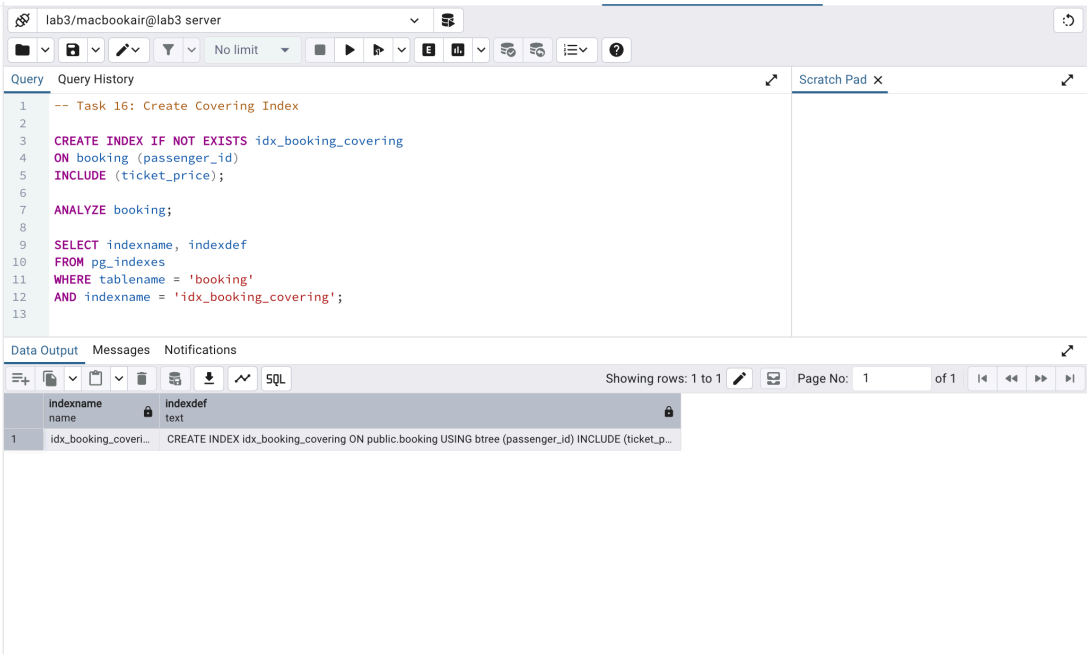

lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 14: City only
EXPLAIN (ANALYZE, BUFFERS)
SELECT *
FROM airport
WHERE city = 'Almaty';
Data Output
Messages
Notifications
Showing rows: 1 to 6
Page No: 1
of 1
QUERY PLAN
text
1 Seq Scan on airport (cost=0.00..1.05 rows=1 width=60) (actual time=0.065..0.067 rows=1.00 loop...
2 Filter: ((city)::text = 'Almaty'::text)
3 Rows Removed by Filter: 3
4 Buffers: shared hit=1
5 Planning Time: 0.166 ms
6 Execution Time: 0.099 ms


lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 14: Country and City
EXPLAIN (ANALYZE, BUFFERS)
SELECT *
FROM airport
WHERE country = 'Kazakhstan'
AND city = 'Almaty';
Data Output
Messages
Notifications
Showing rows: 1 to 6
Page No: 1
of 1
QUERY PLAN
text
1 Seq Scan on airport (cost=0.00..1.06 rows=1 width=60) (actual time=0.076..0.077 rows=1.00 loop...
2 Filter: (((country)::text = 'Kazakhstan'::text) AND ((city)::text = 'Almaty'::text))
3 Rows Removed by Filter: 3
4 Buffers: shared hit=1
5 Planning Time: 0.193 ms
6 Execution Time: 0.110 ms


lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 15: Create Functional Index
CREATE INDEX IF NOT EXISTS idx_passengers_lower_lastname
ON passengers (LOWER(last_name));
ANALYZE passengers;
SELECT indexname, indexdef
FROM pg_indexes
WHERE tablename = 'passengers'
AND indexname = 'idx_passengers_lower_lastname';
Data Output
Messages
Notifications
Showing rows: 1 to 1
Page No: 1
of 1
indexname
name
indexdef
text
1
idx_passengers_lower_lastna...
CREATE INDEX idx_passengers_lower_lastname ON public.passengers USING btree (lower((last_name)::t...


lab3/macbookair@lab3 server
No limit
Query
Query History
Scratch Pad
-- Task 15: With LOWER()
EXPLAIN (ANALYZE, BUFFERS)
SELECT *
FROM passengers
WHERE LOWER(last_name) = 'irnis';
Data Output
Messages
Notifications
Showing rows: 1 to 8
Page No: 1
of 1
QUERY PLAN
text
1 Seq Scan on passengers (cost=0.00..1.14 rows=1 width=69) (actual time=0.034..0.034 rows=0.00 loop...
2 Filter: (lower((last_name)::text) = 'irnis'::text)
3 Rows Removed by Filter: 9
4 Buffers: shared hit=1
5 Planning:
6 Buffers: shared hit=51 read=1
7 Planning Time: 1.159 ms
8 Execution Time: 0.094 ms


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No limit
Query
Query History
Scratch Pad
-- Task 15: Without LOWER()
EXPLAIN (ANALYZE, BUFFERS)
SELECT *
FROM passengers
WHERE last_name = 'Irnis';
Data Output
Messages
Notifications
Showing rows: 1 to 6
Page No: 1
of 1
QUERY PLAN
text
1 Seq Scan on passengers (cost=0.00..1.11 rows=1 width=69) (actual time=0.057..0.057 rows=0 loop...
2 Filter: ((last_name)::text = 'Irnis'::text)
3 Rows Removed by Filter: 9
4 Buffers: shared hit=1
5 Planning Time: 0.203 ms
6 Execution Time: 0.087 ms


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No limit
Query
Query History
Scratch Pad
-- Task 16: Create Covering Index
CREATE INDEX IF NOT EXISTS idx_booking_covering
ON booking (passenger_id)
INCLUDE (ticket_price);
ANALYZE booking;
SELECT indexname, indexdef
FROM pg_indexes
WHERE tablename = 'booking'
AND indexname = 'idx_booking_covering';
Data Output
Messages
Notifications
Showing rows: 1 to 1
Page No: 1
of 1
indexname
name
indexdef
text
1 idx_booking_coveri... CREATE INDEX idx_booking_covering ON public.booking USING btree (passenger_id) INCLUDE (ticket_p...

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No limit

Query | Query History | Scratch Pad

```sql
-- Task 16: Check Index-Only Scan

EXPLAIN (ANALYZE, BUFFERS)
SELECT ticket_price
FROM booking
WHERE passenger_id = 1;
```

Data Output | Messages | Notifications

Showing rows: 1 to 8 | Page No: 1 of 1

	QUERY PLAN text
1	Seq Scan on booking (cost=0.00..1.11 rows=3 width=6) (actual time=0.016..0.018 rows=3.00 loop…
2	Filter: (passenger_id = 1)
3	Rows Removed by Filter: 6
4	Buffers: shared hit=1
5	Planning:
6	Buffers: shared hit=28 read=1 dirtied=3
7	Planning Time: 0.942 ms
8	Execution Time: 0.059 ms

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No limit

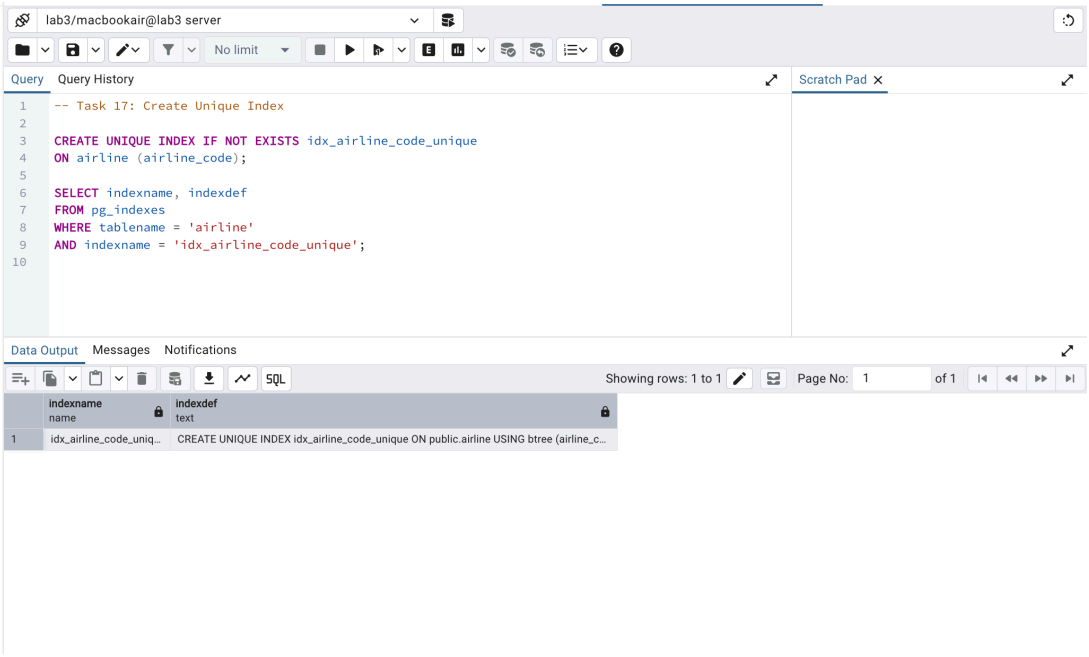
Query | Query History | Scratch Pad

```sql
-- Task 17: Step 1 - Find duplicate airline codes

SELECT airline_code,
       COUNT(*) AS total
FROM airline
GROUP BY airline_code
HAVING COUNT(*) > 1;
```

Data Output | Messages | Notifications

	airline_code character varying (30)	total bigint


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No limit
Query
Query History
Scratch Pad
1 -- Task 17: Create Unique Index
2
3 CREATE UNIQUE INDEX IF NOT EXISTS idx_airline_code_unique
4 ON airline (airline_code);
5
6 SELECT indexname, indexdef
7 FROM pg_indexes
8 WHERE tablename = 'airline'
9 AND indexname = 'idx_airline_code_unique';
10
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 1
Page No: 1
of 1
indexname
name
indexdef
text
1
idx_airline_code_uniq...
CREATE UNIQUE INDEX idx_airline_code_unique ON public.airline USING btree (airline_c...